Finite Element Methods

Lecture Notes, Fall 2027

transcribed, typeset, and annotated for a mechanics reader

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How to read these notes

These are my notes for a graduate finite element course taught out of the mathematics department. The mathematical content is transcribed from lecture; the coloured boxes are mine, added while working through the material.

Every lecture opens with an orange *big picture* box. The lecturer writes down perhaps half of what he says, so that box is the missing half: what this lecture is for, in bullets, and what it buys an engineer. Read it first.

The big picture: what this box is for

The question the lecture answers.

- Three or four bullets — what actually gets built, and why that step is needed rather than the one before it.

Use it for: The concrete thing you can do afterwards that you could not do before.

Intuition: what this box is for

Explains what an object *is*, stripped of formalism — what it measures, why it was invented, and what would break without it.

In mechanics terms

Restates the same statement as a bar, a beam, a heat-conducting slab, or a stiffness matrix — something with units.

Translation

Math department writes	You should hear
$a(u, v)$, bilinear form	internal virtual work; the stiffness operator
V, Hilbert space	the set of displacement fields with finite strain energy
coercivity	the stiffness matrix is positive definite

Where this trips people up

Marks a specific place where the mathematician's phrasing and the engineer's phrasing describe the same thing but sound unrelated.

Recall

Background the lecturer assumed everyone had.

► **Unclear in my notes:** Marks something I could not read or did not follow in my own handwriting — these are the questions to bring to office hours.

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Chapter 1

Lectures

1.1 Why FEM, and the two-point BVP

Lecture 1 — Monday, August 24, 2026 Math 6660

The big picture: Lecture 1

Why does FEM not just solve the differential equation?

- The model problem for the whole course is $-(au')' = f$. Read it as an **axial bar**: $a = EA$ is the stiffness, f the distributed load, u the displacement. Everything later is this equation or a relative of it.
- FEM never attacks the *strong* form (the ODE holding at every point). It rewrites the problem as one *integral* identity, tested against arbitrary virtual displacements. That is the weak form.
- The operator turns out to be **self-adjoint** — which is Betti–Maxwell reciprocity wearing a different hat.
- The space V of admissible displacements is defined by **finite strain energy**, not by smoothness. That choice is what lets kinked, piecewise-linear elements be legal functions.

Use it for: Knowing why every element derivation you have ever seen starts from virtual work rather than from the governing PDE.

Context

Where this course sits relative to the rest of the numerical sequence:

Prerequisites	**The discretisation courses**
5610: interpolation, integration, ODEs	**6660**: FEMs (this course)
5620: numerical linear algebra	6670: FVMs
	5170: FDMs

Why FEM?

(1) Theoretical. The original problem is continuous.

- **FDM**: approximate the solution at a discrete number of points.
- **FEM**: keep the original continuous problem (the PDE), but replace (approximate) the *function spaces*.

All the theory of PDEs carries over to FEM.

Intuition: FDM discretises the equation; FEM discretises the space

This is the single most important sentence of the lecture, and it is the reason the rest of the course looks like functional analysis rather than calculus.
FDM replaces $\mathrm{d}/\mathrm{d}x$ by a difference quotient — it attacks the *operator*, and what survives is a list of nodal values with no meaning in between. FEM leaves the equation alone and instead shrinks the *set of candidate shapes* from "all displacement fields with finite energy" to "all piecewise polynomials on this mesh." The answer is still a function defined everywhere; you can differentiate it, integrate it, and ask for its value between nodes.
That is why an FE result gives you a stress field over the whole part while a finite-difference result gives you numbers at grid points, and it is why FEM inherits the PDE's theory: the problem was never changed, only the space it is posed over.

(2) Practical reasons.

- Can be formulated on an unstructured grid.
- Lots of flexibility in the choice of function spaces.

In mechanics terms

Both bullets are why FEM won in engineering practice and FDM did not. An unstructured grid is what lets you mesh a bracket with a fillet and three bolt holes; a structured finite-difference grid cannot follow that boundary. "Flexibility in the choice of function spaces" is element selection — linear vs. quadratic, Lagrange vs. Hermite, beam vs. shell vs. solid. Choosing an element type *is* choosing V_h.

(3) Note. Much of the theory of FEMs is based on *elliptic* PDE theory. Parabolic PDEs are OK. Hyperbolic PDEs are more challenging.

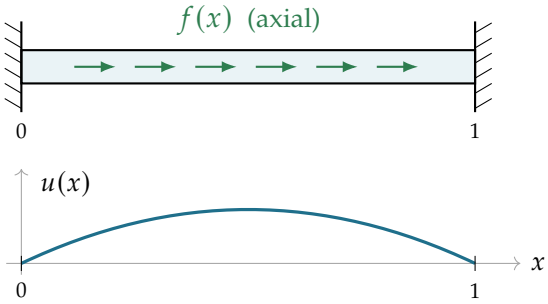
Intuition: why elliptic problems are the friendly case

Elliptic = steady-state and diffusive: static stress analysis, steady heat conduction, incompressible potential flow. Information propagates instantaneously in all directions, the solution is as smooth as the data allows, and — the part that matters for this course — the problem has an associated *energy* that the solution minimises. That energy is what becomes the bilinear form $a(u, v)$, and its positivity is what makes every existence proof work.
Hyperbolic = wave-like: information travels along characteristics at finite speed, discontinuities (shocks, wavefronts) propagate rather than smooth out, and there is no energy functional to minimise. Standard Galerkin FEM has no mechanism to respect a direction of propagation, which is why it produces spurious oscillations on advection-dominated problems and why that case needs separate machinery (SUPG, discontinuous Galerkin). If you have run an explicit crash or blast simulation and wondered why it uses such different technology from a static solve, this is why.

The two-point BVP

The 1D version of Poisson's equation. On the fixed domain $0 < x < 1$:

$$-\big(a(x)\,u'\big)' = f(x), \qquad u(0) = u(1) = 0. \tag{1.1}$$



Where this trips people up: do not draw this as a sagging beam

The obvious picture — downward arrows and a drooping curve — is **wrong**, and the lecturer flags it directly: *"misleading figure; try to think of it as a heat transfer problem rather than a structural mechanics problem."*
Here u is *axial* displacement and f is an *axial* load, so the load arrows point along the bar, and the curve below is a *plot of u against x*, not the shape the bar takes. A sagging picture silently imports bending, which is a fourth-order problem with a different energy ($\int EI(w'')^2$) and different elements — see the H^1 vs H^2 discussion later in this lecture.
The safest reading of (1.1) is steady heat conduction: u temperature, a conductivity, f a heat source. Nothing there tempts you into bending. Use the bar for the *algebra* ($au' = N$, $\int a(u')^2$ = strain energy); use heat conduction for the *picture*.

u := "displacement"
f := "external load" given
a = "modulus of elasticity" given

Assume:

$$0 < a_1 \le a(x) \le a_2 < \infty \qquad \forall\, x \in [0,1]. \tag{1.2}$$

In mechanics terms: the model problem is an axially loaded bar

The lecturer's own naming makes this explicit, so take it literally. Let $a(x) = E(x)A(x)$. Then au' is the internal axial force $N(x) = EA\,\varepsilon$, and (1.1) reads

$$-\frac{\mathrm{d}N}{\mathrm{d}x} = f \qquad \text{i.e.} \qquad \frac{\mathrm{d}N}{\mathrm{d}x} + f = 0,$$

which is nothing but axial equilibrium of a differential slice of bar. The boundary condition $u(0) = u(1) = 0$ pins both ends.
So (1.1) is: *a bar, fixed at both ends, under distributed axial load f, made of a material whose stiffness a may vary along its length.* Every abstract statement in the next four lectures can be checked against this picture.
Read (1.2) the same way: $a_1 > 0$ says the bar is nowhere made of nothing (no zero-stiffness segment, or it would separate freely), and $a_2 < \infty$ says it is nowhere rigid. These two constants will reappear later as the coercivity and continuity constants, and their ratio a_2/a_1 is the material contrast that controls how bad the error bound gets.

► **Unclear in my notes:** Page 3 of my notes writes the boundary condition as $u(0) = u(1) = 1$ at the top, but every subsequent line — and all of Lecture 2 — uses $u(0) = u(1) = 0$. I have transcribed it as 0. Confirm it was a slip.

The operator is self-adjoint

Note that

$$\mathcal{L}(u) = -(au')' \qquad \text{with} \qquad u(0) = u(1) = 0 \tag{1.3}$$

is a linear differential operator, and $\mathcal{L}$ is *self-adjoint*.

To see it, integrate by parts twice:

$$\int_0^1 v\,\mathcal{L}(u)\,\mathrm{d}x = -\int_0^1 v\,(au')'\,\mathrm{d}x$$
$$= \int_0^1 v'\,a\,u'\,\mathrm{d}x - \Big[v\,a\,u'\Big]_{x=0}^{x=1}. \tag{1.4}$$

Assume $u(0) = u(1) = 0$ and $v(0) = v(1) = 0$; the boundary term drops:

$$= \int_0^1 v'\,a\,u'\,\mathrm{d}x$$
$$= -\int_0^1 u\,(av')'\,\mathrm{d}x + \Big[u\,a\,v'\Big]_{x=0}^{x=1}$$
$$= -\int_0^1 u\,(av')'\,\mathrm{d}x = \int_0^1 u\,\mathcal{L}(v)\,\mathrm{d}x. \tag{1.5}$$

$$\boxed{\int_0^1 v\,\mathcal{L}(u)\,\mathrm{d}x = \int_0^1 u\,\mathcal{L}(v)\,\mathrm{d}x} \qquad v = \text{test function} \tag{1.6}$$

This is what it means to be self-adjoint.

► **Unclear in my notes:** My notes write line (1.4) as $\int v\,\mathcal{L}(u) = \int v\,(au')'$, without the minus sign, even though $\mathcal{L}(u) = -(au')'$ two lines above. The very next line *does* carry the sign correctly, so it is a copying slip and I have restored the minus here.

Intuition: self-adjointness is Betti's reciprocal theorem

Equation (1.6) is not a piece of abstract bookkeeping — it is a theorem you were taught in structures, under a different name.

Read $\mathcal{L}(u)$ as "the load distribution that produces displacement field u." Then the left side is *the work done by load system u moving through the displacements of system v*, and the right side is the work done by load system v moving through the displacements of u. Equation (1.6) says these are equal.

That is **Betti–Maxwell reciprocity**: push at A and measure the deflection at B; you get the same number as pushing at B and measuring at A. It is also exactly why the flexibility and stiffness matrices are symmetric.

Note. A self-adjoint differential operator $\mathcal{L}$ is analogous to a symmetric matrix.

Translation

Math department writes	You should hear
$\mathcal{L}$ self-adjoint	$\boldsymbol{K} = \boldsymbol{K}^T$
$\int v\,\mathcal{L}(u) = \int u\,\mathcal{L}(v)$	$\boldsymbol{v}^T\boldsymbol{K}\boldsymbol{u} = \boldsymbol{u}^T\boldsymbol{K}\boldsymbol{v}$
the boundary term $[v\,a\,u']_0^1$	end reactions doing work; killed by the pinned BCs
$a\,u'$	internal axial force $N = EA\,\varepsilon$

Where this trips people up: the boundary term is where BCs actually enter

Watch what happened in (1.4): the essential boundary condition was not imposed on the equation, it was used to *delete the boundary term*. This is the mechanism for the whole subject.
Kill the term by forcing $v = 0$ at the end $\Rightarrow$ that end carries an *essential* (Dirichlet, prescribed-displacement) condition, and it must be built into the space. Kill it instead by prescribing $au' =$ given $\Rightarrow$ that end carries a *natural* (Neumann, prescribed-traction) condition, and it appears in the load vector for free, with nothing imposed on the space.
That is the real content of "essential vs. natural," and it is why in a commercial FE code you constrain displacement DOFs explicitly but simply *apply* a traction and let the formulation absorb it.

Invertibility and regularity

Note. Consider

$$\mathcal{L}(w) = 0, \qquad w(0) = w(1) = 0.$$

The only solution that satisfies $\mathcal{L}$ and the BCs is $w = 0$. Therefore $\mathcal{L}$ is "invertible": for given a and f there exists a unique u.

Intuition: trivial null space = no rigid-body modes

$\mathcal{L}(w) = 0$ with $w = 0$ at both ends asks: *is there a nonzero displacement field that requires no load?* For a bar pinned at both ends, obviously not — any nonzero shape stretches some fibre, which needs force. Release one end and the answer changes: $w =$ const becomes a free translation, the null space is nontrivial, and the problem has no unique solution.
In matrix terms: $\boldsymbol{K}\boldsymbol{w} = \boldsymbol{0} \Rightarrow \boldsymbol{w} = \boldsymbol{0}$ means $\boldsymbol{K}$ is nonsingular. It is the same check as "did I constrain enough DOFs to remove all rigid body motion," which is the first thing a solver complains about when it reports a singular stiffness matrix.

► **Unclear in my notes:** My notes read "there exist a unique u and f" — but f is given data, not something solved for. I have written "for given a and f there exists a unique u." Check the phrasing.

Note (regularity). If $0 < a < \infty$ and f are sufficiently smooth — specifically, if

$$a \in C^1((0,1)) \quad \text{and} \quad f \in C^0((0,1)),$$

then $u \in C^2((0,1))$.

Intuition: the solution is two derivatives smoother than the load

$\mathcal{L}$ takes two derivatives, so solving with $\mathcal{L}^{-1}$ gives two back. This is the *elliptic regularity* pattern, and it is why the elliptic case is the friendly one.
It also tells you where it breaks: the hypothesis needs $a \in C^1$, and a bar made of two glued materials has a *jumping* at the interface. Then $u \notin C^2$, the strong form (1.1) is not satisfied in the classical sense, and there is no classical solution at all — even though the bar plainly has a displacement field. Resolving that is precisely what the variational formulation is for, and Lecture 2 closes the loop on it.

Variational formulation

Multiply our BVP by a test function and integrate over the domain $[0,1]$:

$$-\int_0^1 (au')'\, v \,\mathrm{d}x = \int_0^1 \underbrace{f}_{\text{force}}\, v \,\mathrm{d}x. \tag{1.7}$$

Then integrate by parts:

$$\int_0^1 a\,u'v'\,\mathrm{d}x - \Big[a\,u'v\Big]_0^1 = \int_0^1 f\,v\,\mathrm{d}x. \tag{1.8}$$

Assume v satisfies the BCs:

$$v(0) = v(1) = 0.$$

In particular, define the solution space

$$V = \left\{ v \;:\; \int_0^1 v^2\,\mathrm{d}x < \infty, \quad \int_0^1 \big[v'\big]^2\,\mathrm{d}x < \infty, \quad v(0) = v(1) = 0 \right\}. \tag{1.9}$$

Intuition: V is "every shape the bar could actually take"

Read the three conditions in (1.9) physically, in order:

- $\int v^2 < \infty$ — the displacement is finite in the mean. The bar does not fly off to infinity.
- $\int (v')^2 < \infty$ — **finite strain energy.** Since $v' = \varepsilon$, this integral is (up to the factor $a/2$) the stored elastic energy. This is the condition that does all the work.
- $v(0) = v(1) = 0$ — kinematically admissible: the shape respects the supports.

So V is the set of *kinematically admissible displacement fields with finite strain energy*. Nothing more exotic than that. This space is called $H_0^1(0,1)$ once the course introduces Sobolev notation; when you meet that symbol, substitute this sentence.

What each rung of the derivative ladder is. Every condition defining V is a statement about one of these, and nothing in the list goes past the first derivative:

Object	Mechanics	Units	Where it appears
v	displacement	m	$\int v^2 < \infty$; the BCs $v(0) = v(1) = 0$
v'	strain ε	—	$\int (v')^2 < \infty$: the energy condition
av'	stress σ (times area)	Pa	the flux; continuous across a material joint
$(av')'$	net force per length	N/m	*only* in the strong form $-(av')' = f$

Because $0 < a_1 \le a \le a_2 < \infty$, the middle two rows say the same thing: $a_1^2 \int (v')^2 \le \int \sigma^2 \le a_2^2 \int (v')^2$, so "finite strain energy", "v' square integrable" and "σ square integrable" are interchangeable. That two-sided bound on a is exactly what keeps them from coming apart.

Why v'' is not required. Two independent reasons, and it is worth holding both:

(i) **The energy never contains it.** Strain energy is $\frac{1}{2}\int a(v')^2$ — first derivative only. A condition on v'' would be restricting something the physics does not charge for.

(ii) **Integration by parts removed it.** The strong form demands u''; (1.8) moved one derivative onto v, leaving u' and v'. After that step v'' has left the problem entirely, and u', v' appear *only inside integrals* — never evaluated at a point.

Hence what V does *not* require: v'' need not exist, and v' need not exist in the classical sense everywhere. A hat function — kinked, not differentiable at its peak — is in V, because that single bad point has zero length and contributes nothing to $\int (v')^2$. That is deliberate: the space was designed to be exactly big enough to hold the piecewise-linear shapes finite elements are built from.

The whole distinction comes down to **kink versus jump** (Figure 1.1), and it is worth learning once because the same two words recur throughout the course.

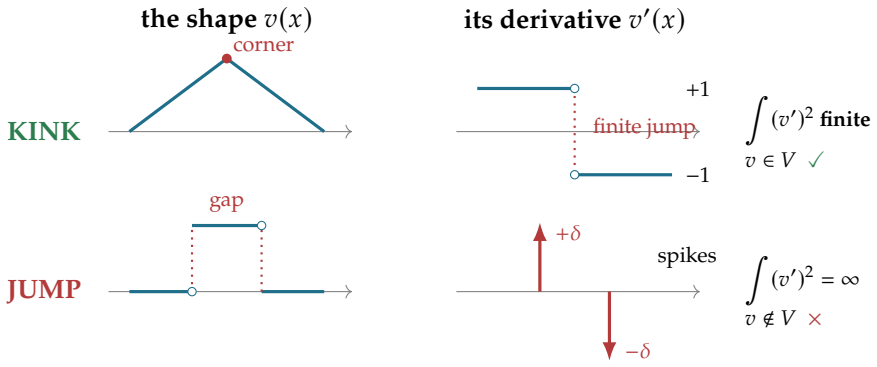


Figure 1.1: **Kink versus jump.** A *kink* is a break in the slope: the shape stays connected, so the strain merely jumps between two finite values and the energy $\int (v')^2$ stays finite. A *jump* is a break in the shape itself: the strain becomes a delta spike and the energy is infinite. Mechanically, a kink is a **material interface** (steel glued to aluminium: displacement continuous, strain jumps because E jumps) — ordinary and everywhere. A jump is a **broken bar**. V admits the first and excludes the second, automatically, with no continuity rule written anywhere.

In mechanics terms: the same rule for beams: why they need rotational DOFs

The rule generalises: *the highest derivative appearing in the energy fixes the space, which fixes the continuity the elements must have.* Working it out for a beam explains an element-selection rule you already follow.

For a beam the derivative ladder is shifted up by one rung, because bending energy lives in *curvature* rather than in stretch:

	Bar (this course, for now)	**Beam (Euler–Bernoulli)**
field	v = axial displacement	w = transverse deflection
first derivative	v' = strain ε	w' = slope θ
second derivative	—	w'' = curvature κ
internal measure	$\sigma = Ev'$	$M = EI\,w''$ (bending moment)
energy	$\frac{1}{2}\int EA\,(v')^2$	$\frac{1}{2}\int EI\,(w'')^2$
equilibrium	$-(EA\,v')' = f$ (2nd order)	$(EI\,w'')'' = q$ (4th order)
space	H^1: $\int (v')^2 < \infty$	H^2: $\int (w'')^2 < \infty$
must stay continuous	v	w *and* w'
forbidden	jump in v (a break)	jump in w' (a hinge)
element	C^0, DOF $\{w\}$	C^1, DOF $\{w, \theta\}$
basis	linear hat (Lagrange)	cubic Hermite

Now run the Figure 1.1 argument one rung higher. For the bar, a *jump in* v gives a delta in v' and infinite energy, so it is forbidden; a *kink in* v is fine. For the beam the energy contains w'', so the thing that must not jump is w':

$$\begin{aligned} &\text{bar:} \quad \text{jump in } v \;\Rightarrow\; \delta \text{ in } v' \;\Rightarrow\; \textstyle\int (v')^2 = \infty \;\times \\ &\text{beam: jump in } w' \Rightarrow \delta \text{ in } w'' \Rightarrow \textstyle\int (w'')^2 = \infty \;\times \end{aligned}$$

What a slope jump physically is: a hinge. A kink in a beam's deflected shape means the two sides meet at an angle. The curvature there is infinite — you have turned through a finite angle over zero length — so the bending energy $\frac{1}{2}\int EI\kappa^2$ is infinite. A genuine continuous beam cannot do this. (A real hinge *can*, but only because it has $EI = 0$ there, so it stores no energy; that is a different structure, and you model it by deliberately releasing the rotational DOF.)

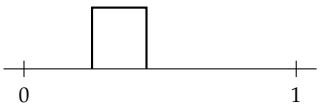
The consequence for elements. Two adjacent bar elements sharing a node share one number, the displacement — and that alone makes v continuous, so C^0 suffices. Two adjacent beam elements must agree on *both* the deflection and the rotation at the shared node, or a hinge appears at every node of your mesh. That is why a beam element carries $\{w_1, \theta_1, w_2, \theta_2\}$ — four DOFs, hence a cubic (C^1 Hermite) shape function — while a bar element carries only $\{w_1, w_2\}$.
So when a solver gives a beam or shell node rotational degrees of freedom and a truss or solid node none, that is not a modelling convention. It is forced by which derivative appears in the energy. Plates and shells are the 2D version of the same statement.

Where this trips people up: why the trial space suddenly needs only *one* derivative

Compare (1.7) with (1.8). Before integrating by parts, u is hit with two derivatives. After, the derivatives are shared: one on u, one on v. Nothing was approximated — the two lines are equal — but the smoothness demanded of u has dropped from C^2 to "u' square integrable."
That trade is the entire reason the weak form exists, and it is why FEM can use C^0 elements to solve a second-order PDE. In ME language: the principle of virtual work never asked for the second derivative of displacement in the first place. It only ever needed strains.

Note. Functions in V are continuous.

(1) $\int_0^1 v^2 \, \mathrm{d}x < \infty$ *admits* discontinuous functions:



(2) $\int_0^1 \left[v'\right]^2 \mathrm{d}x$ does *not* allow discontinuities in V.

Intuition: a jump in displacement means the bar is broken

It is condition (2), not (1), that forces continuity — and the mechanical reading is immediate. If v jumps at a point, the bar has separated there: a gap has opened. The strain at that point is infinite (a finite displacement change over zero length), so the strain energy is infinite.
V therefore excludes cracked configurations automatically, without anyone writing down a continuity requirement. Requiring finite energy *implies* C^0. Lecture 2 verifies this by explicit computation with a step function, and the answer comes out $+\infty$ exactly as this picture predicts.
This also explains the C^0 compatibility requirement you know from element selection: adjacent elements must share displacement along their common edge, or the assembled field has a jump there and the energy is meaningless.

Summary: what the strong and weak forms each demand

Where this trips people up: the weak form needs H^1, *not* C^1

An easy slip is to say "strong needs C^2, weak needs C^1." The second half is wrong, and the difference is the whole point:

$$C^1: \ u' \text{ exists } \textit{and is continuous} \quad \Longrightarrow \textbf{ no kink in } u$$
$$H^1: \ \int (u')^2 \, dx < \infty, \ u' \text{ may jump} \quad \Longrightarrow \textbf{ kink in } u \textbf{ allowed}$$

H^1 is *weaker* than C^1:

$$C^2 \subset C^1 \subset H^1 \subset C^0.$$

Had the weak form demanded C^1, hat functions would be illegal and piecewise-linear FEM could not exist.

	Strong form ($u \in C^2((0,1))$)	**Weak form** ($u \in V = H^1_0$)
demands on u	u, u', u'' all continuous	u continuous, $\int (u')^2 < \infty$
u' may jump?	no ×	**yes** ✓
u'' required?	yes	never appears
equation holds	at every point	in an integral (averaged) sense

What the weak form buys.

	strong	**weak**
material interface (a jumps)	no solution ×	fine ✓
point load, re-entrant corner	no solution ×	fine ✓
hat functions	illegal ×	legal ✓ (FEM exists only here)
element must compute	u''	only u' (the $\boldsymbol{B}$ matrix)
natural BC (traction)	imposed by hand	free — appears in the load vector
existence/uniqueness	no theory	Lax–Milgram (H^1 is Hilbert)
sparsity	—	small-support kinked basis ⇒ banded $\boldsymbol{K}$

What it costs.

- The PDE cannot be checked at a point — only weighted averages of it.
- Strain and stress come out *discontinuous* across element boundaries; contour plots need smoothing before presentation.
- Derivatives are one order less accurate than the displacement, and are best sampled at Gauss points rather than nodes.
- Conceptual overhead: Sobolev spaces and weak derivatives, i.e. the first two lectures of this course.

Intuition: the trade in one line

strong: demand a lot of u ⇒ pointwise information, but usually *no solution*

weak: demand little of u ⇒ a solution always exists, but only in energy terms

Engineering keeps the weak form because "no solution at all" is a worse problem than "stress needs smoothing before you plot it."

1.2 Weak solutions: the three claims

Lecture 2 Math 6660 — continues in the same OneNote export, from p. 7

The big picture: Lecture 2

Is the weak solution the same thing as the real solution?

- A weak solution is allowed to have a **kink**; a strong solution is not. The example: a bar whose EA jumps at a point — physically fine, but u'' does not exist there.
- **Claim 1:** the weak solution is unique. One problem, one answer.
- **Claim 2:** strong $\Rightarrow$ weak. Nothing is invented by going to the integral form.
- **Claim 3:** weak + enough smoothness $\Rightarrow$ strong. Nothing is lost either.

Use it for: Trusting that the answer a virtual-work formulation produces is *the* answer, not an approximation to some neighbouring problem.

Setup, restated

Two-point BVP. On the fixed domain $0 < x < 1$:

$$-(au')' = f, \qquad u(0) = u(1) = 0, \qquad \text{find } u \in C^2((0,1)). \tag{1.10}$$

Variational formulation.

$$V = \left\{ v \;:\; \int_0^1 v^2 \,\mathrm{d}x < \infty, \quad \int_0^1 (v')^2 \,\mathrm{d}x < \infty, \quad v(0) = v(1) = 0 \right\} \qquad \text{(defining a space)} \tag{1.11}$$

Note. Functions in V are continuous $\big(v \in C^0((0,1))\big)$.

► **Unclear in my notes:** Page 7 of my notes writes the first condition as $\int_0^1 v^2 \,\mathrm{d}x < 0$. That must be $< \infty$ — the integrand is a square, so < 0 is impossible. Transcribed as $< \infty$.

Example of a discontinuity

Let

$$H(x) = \begin{cases} 1, & \tfrac{1}{4} < x < \tfrac{3}{4}, \\ 0, & \text{otherwise}, \end{cases} \tag{1.12}$$

which is not continuous: $H \notin C^0((0,1))$, but $H \in C^{-1}((0,1))$ — it has a finite jump.

So

$$\int_0^1 H^2 \,\mathrm{d}x = \int_{1/4}^{3/4} \mathrm{d}x = \tfrac{1}{2} \;<\; \infty. \tag{1.13}$$

The first condition in (1.11) is satisfied. Now the derivative:

$$H'(x) = \delta\big(x - \tfrac{1}{4}\big) - \delta\big(x - \tfrac{3}{4}\big) \;=\; \delta_{1/4} - \delta_{3/4}, \tag{1.14}$$

writing $\delta_{1/4} \equiv \delta(x - \frac{1}{4})$ and $\delta_{3/4} \equiv \delta(x - \frac{3}{4})$ — the same objects, shorter names. Each jump contributes one delta, sited at the jump, with coefficient equal to the *signed* jump size:

at	H goes	size	contributes
$x = \frac{1}{4}$	$0 \to 1$	$+1$	$+\,\delta_{1/4}$
$x = \frac{3}{4}$	$1 \to 0$	-1	$-\,\delta_{3/4}$

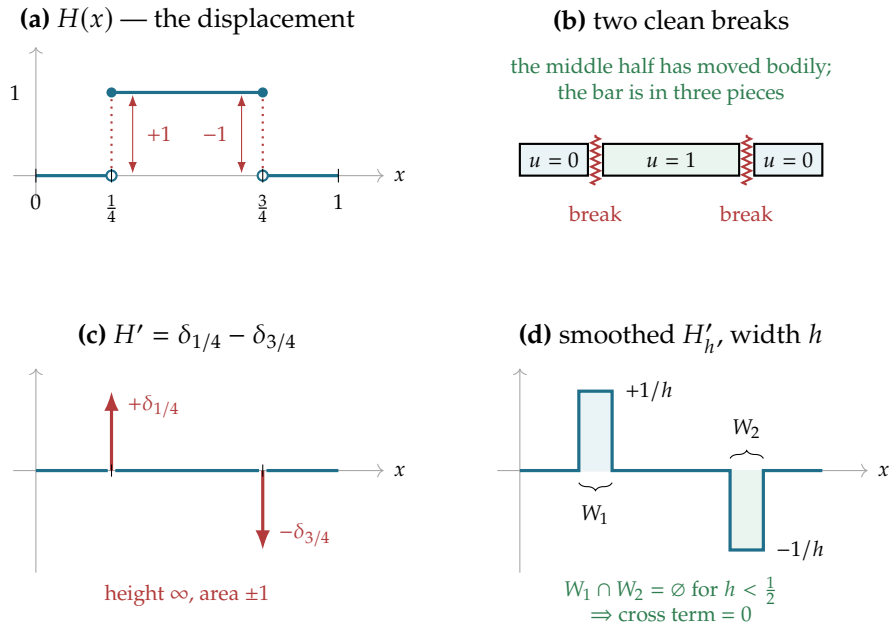


Figure 1.2: The step H: (a) the displacement field, jumping +1 then −1; (b) the same thing as a bar — three pieces, two breaks; (c) its derivative, a pair of deltas; (d) the smoothed version used for the honest calculation.

Recall: δ properties

For $\delta(x - x_0)$:

$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} \delta(x - x_0)\,dx = 1 \qquad \text{for any } \varepsilon_1, \varepsilon_2 > 0, \tag{1.15}$$

$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} f(x)\,\delta(x - x_0)\,dx = f(x_0), \qquad f \text{ a continuous function.} \tag{1.16}$$

Now expand the square — three terms, and *all three* must be accounted for:

$$\begin{aligned}\int_0^1 (H')^2\,dx &= \int_0^1 \big(\delta_{1/4} - \delta_{3/4}\big)^2\,dx \\ &= \int_0^1 \underbrace{\delta_{1/4}^2}_{\text{(i)}}\,dx \;-\; 2\int_0^1 \underbrace{\delta_{1/4}\,\delta_{3/4}}_{\text{(ii)}}\,dx \;+\; \int_0^1 \underbrace{\delta_{3/4}^2}_{\text{(iii)}}\,dx.\end{aligned} \tag{1.17}$$

An arrow is then drawn from term (i) down to the assertion

$$\int_0^1 \delta\big(x - \tfrac{1}{4}\big)\,\delta\big(x - \tfrac{1}{4}\big)\,dx = +\infty, \tag{1.18}$$

and the conclusion $H \notin V$ follows immediately.

► **Unclear in my notes:** Two gaps here, both in the lecture as written. **(a)** No justification is offered for (1.18) — and (1.16) cannot supply one, since it requires f to be a *continuous function* while δ is not a function at all, so $f = \delta$ is not a legal substitution. **(b)** Only term (i) is touched; (ii) and (iii) are never addressed. The *conclusion* (1.18) is correct — the derivation below supplies what is missing.

The honest computation. Replace each jump by a ramp of width h. Then H_h' is an ordinary piecewise-constant *function*:

$$H_h'(x) = \begin{cases} +1/h, & x \in W_1 = \left(\frac{1}{4} - \frac{h}{2},\ \frac{1}{4} + \frac{h}{2}\right), \\ -1/h, & x \in W_2 = \left(\frac{3}{4} - \frac{h}{2},\ \frac{3}{4} + \frac{h}{2}\right), \\ 0, & \text{elsewhere,} \end{cases} \qquad \text{and for } h < \tfrac{1}{2},\ W_1 \cap W_2 = \varnothing. \tag{1.19}$$

Now every term of (1.17) can be evaluated with no distributions at all:

	term	**smoothed integrand**	**integral**
(i)	$\delta_{1/4}^2$	$(1/h)^2$ on W_1, else 0	$\frac{1}{h^2} \cdot h = \frac{1}{h}$
(ii)	$-2\,\delta_{1/4}\delta_{3/4}$	0 everywhere ($W_1 \cap W_2 = \varnothing$)	0
(iii)	$\delta_{3/4}^2$	$(-1/h)^2$ on W_2, else 0	$\frac{1}{h^2} \cdot h = \frac{1}{h}$
	total		$\frac{1}{h} + 0 + \frac{1}{h} = \frac{2}{h} \longrightarrow \infty$

Note (ii): the cross term vanishes for a *rigorous* reason, not a hand-wave — the two windows are disjoint, so wherever one factor is nonzero the other is zero. **Deltas at different places never interact.** And (iii) contributes $+1/h$, not $-1/h$: the sign is destroyed by the square, so the two spikes *add*.

Intuition: finite area, infinite squared area

No sampling property is needed anywhere. The one fact is

$$\int \delta \, dx = 1 \quad \text{(finite \textbf{area})}, \qquad \text{but} \qquad \int \delta^2 \, dx = \infty.$$

A delta has finite area but **infinite height**; squaring the height beats the shrinking width. Seen in the smoothed numbers, with $H = 1/h$ and $W = h$:

$$\underbrace{H \cdot W = \tfrac{1}{h} \cdot h = 1}_{\text{plain: the } h\text{'s cancel}} \qquad \text{vs.} \qquad \underbrace{H^2 \cdot W = \tfrac{1}{h^2} \cdot h = \tfrac{1}{h}}_{\text{squared: one } h \text{ survives below}}$$

Recall: the plain integral passes — as always

Run (1.19) through the *unsquared* test as a control:

$$\int_0^1 H_h' \, dx = \underbrace{\left(+\tfrac{1}{h}\right) h}_{W_1} + \underbrace{\left(-\tfrac{1}{h}\right) h}_{W_2} = 1 - 1 = 0 \ = \ H(1) - H(0) \quad \checkmark \text{ (FTC)}.$$

Finite for every h, and independent of h. So the plain condition would *admit* this broken bar. Only the square rejects it.

Therefore

$$\boxed{H(x) \notin V} \qquad \text{because } H' \text{ is not square integrable.} \tag{1.20}$$

In mechanics terms: the calculation behind "the bar is broken"

H is a bar whose middle half has been translated bodily by 1 — two clean breaks, at $x = \frac{1}{4}$ and $x = \frac{3}{4}$. Its "strain" H' is a pair of deltas: infinite strain at the two break points, so

$\int (H')^2 = +\infty$ and H is excluded from V.
Note *which* test it passes and which it fails:

test	value	verdict
$\int H^2 \, dx$	$\frac{1}{2}$	finite ✓ — the displacement is perfectly reasonable
$\int H' \, dx$	0	finite ✓ — the plain derivative test too
$\int (H')^2 \, dx$	∞	× — **only the energy condition rejects it**

That is the precise sense in which "finite strain energy" is a stronger and more physical requirement than "bounded".

Where this trips people up: use the smoothing version in a proof

δ^2 is genuinely undefined — the product of two distributions has no meaning in general, so (1.18) cannot be justified as it stands. If a step like it appears in your own work, write the smoothed version (1.19) and take $h \to 0$: same conclusion, every object an ordinary function.

The variational problem

$$\text{Find } u \in V \text{ such that } \int_0^1 a\, u'v' \, dx = \int_0^1 f v \, dx \qquad \text{for all } v \in V. \tag{1.21}$$

Note. u is the *trial* function; v are the *test* functions. In this case u and v come from the same function space, V.

Intuition: this is the principle of virtual work

Read it term by term with $a = EA$, $u' = \varepsilon$, $v = \delta u$, $v' = \delta\varepsilon$:

$$\underbrace{\int_0^1 \underbrace{au'}_{\sigma A}\,\underbrace{v'}_{\delta\varepsilon}\, dx}_{\text{internal virtual work}} = \underbrace{\int_0^1 f\,\underbrace{v}_{\delta u}\, dx}_{\text{external virtual work}}$$

The test function is the virtual displacement. "For all $v \in V$" is "for every kinematically admissible virtual displacement." The whole statement is $\delta W_{\text{int}} = \delta W_{\text{ext}}$ — equilibrium. You have used this since undergraduate statics. What the math department adds is not the equation but the precision about *which space* u and v live in, and that turns out to be the thing that makes the theory work.

Where this trips people up: trial = test only because the BCs are homogeneous

The note "u and v come from the same function space" is true *here* and not in general. It holds because $u(0) = u(1) = 0$: the prescribed displacement happens to be zero.
With a nonhomogeneous essential BC, say $u(0) = g \neq 0$, the trial function must satisfy $u(0) = g$ while the test function must satisfy $v(0) = 0$ — the virtual displacement cannot violate the support. So the trial set is an affine shift of the test space, not the same space, and it is not even a vector space. The standard fix is to write $u = \tilde{u} + u_g$ with $\tilde{u} \in V$, which is exactly what a solver does internally when it eliminates prescribed DOFs and moves their contribution to the right-hand side.

Q: How are (2-pt BVP) and (variational problem) related?

- BVP requires $u \in C^2((0,1))$.
- Variational problem only needs $u \in V$.

Intuition: the question the next three claims answer

Two different problems are now on the table, and they demand different amounts of smoothness. The natural worry is whether we have quietly changed the problem. Claims 1–3 settle it:

- **Claim 1**: the variational problem has at most one solution.
- **Claim 2**: every BVP solution solves the variational problem (the weak form is not *more* restrictive).
- **Claim 3**: under extra smoothness, every variational solution solves the BVP (a partial converse).

Claim 3 needs hypotheses that Claim 2 does not. That gap is not a defect of the proof — it is where *weak solutions* live, and the final note of the lecture names it.

Claim 1: uniqueness

Proposition 1.2.1. *If the variational problem has a solution, it is unique.*

Proof (by contradiction). Assume there exist 2 different solutions, u and U:

$$\int_0^1 a u' v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \Big| \qquad \int_0^1 a U' v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x.$$

Then subtract:

$$\Rightarrow \quad \int_0^1 a\,(u-U)'\,v' \,\mathrm{d}x = 0. \tag{1.22}$$

Take $v = u - U \in V$. Now

$$\int_0^1 a\Big((u-U)'\Big)^2 \,\mathrm{d}x = 0 \quad \Longrightarrow \quad (u-U)' = 0 \ \text{ on } (0,1) \quad \Longrightarrow \quad u = U + c. \tag{1.23}$$

But by the BCs, $c = 0$, so $u = U$. □

Intuition: a zero-energy difference is a rigid body motion

Follow (1.23) mechanically. Suppose two different displacement fields both equilibrate the same load. Their difference $u - U$ is then a shape that stores *zero* strain energy — the integrand is $a \cdot (\text{strain})^2$ with $a \geq a_1 > 0$, so the only way the integral vanishes is if the strain vanishes everywhere.
Zero strain everywhere means the difference is a rigid body motion: $u - U = c$, a uniform translation. And the supports forbid that, so $c = 0$.
This is the continuum version of "$\boldsymbol{K}$ positive definite $\Rightarrow \boldsymbol{KU} = \boldsymbol{F}$ has a unique solution," and note precisely where each hypothesis is spent: $a \geq a_1 > 0$ gives *positive* definiteness (energy is zero only for zero strain), and the boundary conditions remove the rigid-body null space. Drop either one and uniqueness fails — which is exactly what a solver tells you when it reports a singular stiffness matrix.

Remark 1.2.2. The lecturer labels this "by contradiction," but no contradiction is actually reached — the argument derives $u = U$ directly. It is a direct uniqueness proof.

Claim 2: BVP ⇒ variational

Proposition 1.2.3. *A solution to the BVP is a solution to the variational problem.*

Proof. The BVP has a unique solution $u \in C^2((0,1))$ with $u(0) = u(1) = 0$. If $u \in C^2((0,1))$ and satisfies the BCs, then $u \in V$. The BVP and the variational problem are then equivalent via integration by parts. □

Intuition: the easy direction, and why it is easy

C^2 functions on a bounded interval have bounded derivatives, so $\int (u')^2$ is certainly finite — a classical solution automatically has finite strain energy. Add the BCs and it lands in V. Then (1.7)–(1.8) run forward with no extra hypotheses.
The direction that costs something is the other one: going from "the integral identity holds for every v" back to "the differential equation holds pointwise." That is Claim 3, and it needs both extra regularity *and* a genuine argument.

Claim 3: variational ⇒ BVP

Proposition 1.2.4. *If $f \in C^0((0,1))$, $a \in C^1((0,1))$, and $0 < a_1 \le a(x) \le a_2 < \infty$ for all $x \in (0,1)$, then a solution to the variational problem is a solution to the BVP.*

Proof. Assume for the moment that $u \in C^2((0,1))$. From (1.21),

$$\int_0^1 \big(au'v' - fv\big)\,\mathrm{d}x = 0. \tag{1.24}$$

Reverse integration by parts:

$$\int_0^1 \underbrace{\Big[(au')' + f\Big]}_{S(x)} v\,\mathrm{d}x = 0. \tag{1.25}$$

Goal: show $S(x) = 0$ for all $x \in (0,1)$.

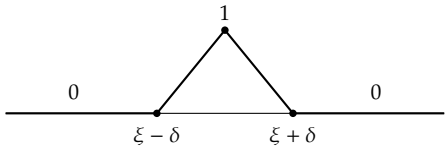
(a) Assume, for contradiction, that

$$S(\xi) = \Big[(au')' + f\Big]\Big|_{x=\xi} > 0 \qquad \text{at some point } \xi \in (0,1).$$

Note that $S(x)$ is continuous in x. By continuity, $S(x) > 0$ for $x \in [\xi - \delta,\ \xi + \delta]$ for some $\delta > 0$.

Choose v to be the following "hat" function:

$v =$



But then

$$\int_0^1 S(x)\,v(x)\,\mathrm{d}x > 0,$$

which is a contradiction with (1.25). Therefore $S(x)$ cannot be positive in $(0,1)$.

(b) Redo this argument with $S(\xi) < 0 \Rightarrow S(x)$ cannot be negative in $(0,1)$.

Hence $S \equiv 0$, which is (1.10). □

Intuition: the hat function is not a coincidence

Two things are worth pausing on here.

First, the logic. This is the *fundamental lemma of the calculus of variations*: if $\int Sv = 0$ for *every* admissible v, then $S = 0$. The mechanical reading is the converse of virtual work — if a leftover force distribution S does zero virtual work against every admissible virtual displacement, there is no leftover force. The proof makes that concrete by picking a virtual displacement that pokes the bar *only* where S is claimed to be positive, so the work integral cannot cancel.

Second, the choice of v. The bump used to localise is the hat function — a piecewise-linear tent, one at ξ, zero outside $[\xi - \delta, \xi + \delta]$. That is precisely the linear finite element basis function φ_i you will build V_h from in a few lectures, and precisely why the stiffness matrix comes out sparse (each hat overlaps only its neighbours).

So the object introduced here as a proof device is the same object that becomes the computational method. Worth noticing now: it is also the cleanest evidence that V was built to contain kinked functions, since a hat has no second derivative at its peak and belongs to V anyway.

Where this trips people up: the proof assumes what it is trying to conclude

The proof opens with "assume for the moment that $u \in C^2(0,1)$" — but $u \in C^2$ is essentially the conclusion of Proposition 1.2.4. As written, the argument shows: *if* a variational solution happens to be C^2, *then* it satisfies the BVP pointwise.

Closing that gap — proving the regularity rather than assuming it — is a separate theorem (elliptic regularity), and the hypotheses $f \in C^0$, $a \in C^1$ are exactly what it needs. The lecturer's "for the moment" signals the debt. The final note below is where it gets paid.

When can we expect $u \in C^2((0,1))$?

$$(\star) \qquad f \in C^0((0,1)), \quad a \in C^1((0,1)), \quad \text{and} \quad 0 < a_1 \le a(x) \le a_2 < \infty \quad \forall\, x \in (0,1). \tag{1.26}$$

Note.

① If $(\star)$ is satisfied: u is a *strong* solution of the 2-pt BVP, and

$$\text{(2-pt BVP)} \;\equiv\; \text{(variational problem)}.$$

② If $(\star)$ is *not* satisfied, the 2-pt BVP will not have a solution, but the variational problem may still have one. Then u (if it exists) is a *weak* solution of the 2-pt BVP.

Intuition: case ② is the whole reason for the course

This is the payoff, and it is worth stating in engineering terms because it describes a situation you have already simulated.

Take a bar made of two materials — steel for $x < \frac{1}{2}$, aluminium for $x > \frac{1}{2}$. Then $a(x)$ *jumps* at the interface, so $a \notin C^1$ and $(\star)$ fails. Consequently:

- The strong form has **no solution**. Not "a hard one" — none. u'' does not exist at the interface, so $-(au')' = f$ is meaningless there.
- The variational problem is **completely fine**. It never asked for u''. It asks for finite strain energy, and a bimaterial bar plainly has that.

The physically correct answer — displacement continuous, internal force au' continuous, strain u' jumping at the interface — is a weak solution and not a strong one.

So the weak formulation is not a relaxation adopted for convenience. It is the formulation

under which the problem you actually want to solve *has* an answer. Every real FE model with more than one material, a contact interface, or a point load lives in case ②.

Where this trips people up: "will not have a solution" is slightly too strong

(⋆) is a *sufficient* condition for $u \in C^2$, not a necessary one, so its failure does not by itself prove that no classical solution exists — one might survive by luck. The accurate statement is that the BVP *need not* have a classical solution when (⋆) fails, while the variational problem still can. The bimaterial bar is a case where it genuinely does fail.

Translation: strong vs. weak

Math department writes	You should hear
strong solution	the classical one: $u \in C^2$, the PDE holds at every point
weak solution	the physical one: finite strain energy, equilibrium holds in the virtual-work sense
(⋆) fails	material interface, point load, re-entrant corner, contact
u solves the variational problem	the structure is in equilibrium against every admissible virtual displacement

Intuition: what the course has actually set up

Standing back from Lectures 1–2, a chain has been built and it is worth naming, because everything later hangs off it:

$$\underbrace{-(au')' = f}_{\text{strong form}} \longrightarrow \underbrace{\int au'v' = \int fv \quad \forall v \in V}_{\text{weak form}} \longrightarrow \underbrace{\text{same, with } V \text{ replaced by } V_h}_{\text{Galerkin / FEM}}$$

The first arrow is what these two lectures did: it costs nothing (Claims 2 and 3) and buys a problem that is solvable in realistic situations. The second arrow has not happened yet — it is where approximation enters, and where $\int au'v'$ becomes the stiffness matrix $K_{ij} = \int a\varphi_j'\varphi_i'$.

Keep the two arrows separate. "Weak" refers to the first and involves no approximation whatsoever; the error you eventually measure comes entirely from the second.

1.3 Energy minimisation and the first finite element method

Lecture 3 — Thursday, August 27, 2026 Math 6660 — OneNote export `3rrd.pdf`, from p. 16

The big picture: Lecture 3

Where does "minimum potential energy" come from, and how does it turn into a computer program?

- Define $F(w)$ = strain energy − work done by the load. Minimising F over V is *exactly* the weak form — proved, not asserted.
- So the two derivations every FEM textbook gives, virtual work and energy minimisa-

tion, are one theorem stated twice.

- Then the actual method: replace the infinite-dimensional V by a finite-dimensional V_h of piecewise linear functions. That is **CG(1)**.
- **Hat functions**: the basis in which each unknown coefficient *is* a nodal displacement.

Use it for: Seeing where the discretisation actually enters — it is one step, restricting the space, and nothing else about the problem changes.

Setup, restated

2-pt BVP.

$$-(au')' = f \ \text{ on } (0,1), \qquad u(0) = u(1) = 0, \tag{1.27}$$

with $f \in C^0((0,1))$, $a \in C^1((0,1))$, and $0 < a_1 \le a(x) \le a_2 < \infty$ for all $x \in [0,1]$.

Variational problem. Find $u \in V$ such that

$$\int_0^1 a\, u'v'\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x \qquad \forall\, v \in V, \tag{1.28}$$

where $V = \{v : \int v^2\,\mathrm{d}x < \infty,\ \int (v')^2\,\mathrm{d}x < \infty,\ v(0) = v(1) = 0\}$.

► **Unclear in my notes:** Page 16 writes the coefficient bound as $0 \le a_1 < a(x) \le a_2$, but Lectures 1 and 2 both have $0 < a_1 \le a(x) \le a_2$. The difference matters: if $a_1 = 0$ were allowed, the ε^2 coefficient in Claim ② below could vanish and the minimiser would stop being unique. I have used the Lecture 1 version.

The "energy" minimisation problem

Define the following energy functional:

$$F(w) \;=\; \underbrace{\frac{1}{2}\int_0^1 a\,(w')^2\,\mathrm{d}x}_{\text{internal energy}} \;-\; \underbrace{\int_0^1 f\,w\,\mathrm{d}x}_{\text{potential energy}} \qquad w \text{ is a function.} \tag{1.29}$$

In mechanics terms: F is the total potential energy Π

This is the most familiar object in the course so far, under a name you have used since undergraduate structures. With $a = EA$ and $w' = \varepsilon$:

$$F(w) \;=\; \underbrace{\tfrac{1}{2}\int_0^1 EA\,\varepsilon^2\,\mathrm{d}x}_{\text{strain energy } U} \;-\; \underbrace{\int_0^1 f\,w\,\mathrm{d}x}_{\substack{\text{work done by the applied}\\ \text{load through } w}} \;=\; U - W \;=\; \Pi.$$

So the Min Problem below is the **principle of minimum total potential energy**: of all kinematically admissible displacement fields, the one that actually occurs is the one that minimises Π. That is the principle behind Rayleigh–Ritz, and — as this lecture proves — behind FEM.

Your own margin note "not total energy" is exactly right and worth keeping: F is the total *potential* energy. There is no kinetic term because the problem is static.

Where this trips people up: "kinetic energy − potential energy" is the wrong label

My notes annotate (1.29) with "kinetic energy − potential energy". That label belongs to a different principle and should not be attached here.

$L = T - V$ (kinetic minus potential) is the *Lagrangian*, and the principle that makes it stationary is Hamilton's principle, which governs *dynamics*. Equation (1.29) is static: the first term is $\frac{1}{2}\int EA\varepsilon^2$, which is **strain energy**, not kinetic energy — there is no velocity anywhere in this problem.

The reason the analogy gets drawn is structural, and it is worth seeing clearly: both functionals are "a positive quadratic term minus a linear term," and in both cases the physical state makes the functional stationary. That shared shape is real. But the physical content of the quadratic term is different, and conflating them will cause trouble later when the course reaches time-dependent problems, where a genuine kinetic term $\frac{1}{2}\int \rho\dot{u}^2$ appears *alongside* the strain energy.

Min problem. Find $u \in V$ such that

$$F(u) \le F(w) \qquad \forall\, w \in V. \tag{1.30}$$

Claim: the Min Problem and the variational problem are equivalent

Theorem 1.3.1. *The minimisation problem* (1.30) *and the variational problem* (1.28) *are equivalent.*

① **(variational problem) ⇒ (Min Prob).** Assume $u \in V$ satisfies $\int_0^1 au'v'\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x$ for all $v \in V$. Let $w \in V$, and set $w = u + v$. Then

$$\begin{aligned} F(w) &= F(u+v) \\ &= \frac{1}{2}\int_0^1 a\,(u'+v')^2\,\mathrm{d}x - \int_0^1 fu\,\mathrm{d}x - \int_0^1 fv\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 a(u')^2\,\mathrm{d}x + \underbrace{\int_0^1 au'v'\,\mathrm{d}x}_{\text{(A)}} + \frac{1}{2}\int_0^1 a(v')^2\,\mathrm{d}x - \int_0^1 fu\,\mathrm{d}x - \underbrace{\int_0^1 fv\,\mathrm{d}x}_{\text{(B)}}. \end{aligned} \tag{1.31}$$

Terms (A) and (B) cancel, since (A) − (B) = 0 is exactly the variational problem. What remains is

$$\boxed{F(w) = F(u) + \frac{1}{2}\int_0^1 a\,(v')^2\,\mathrm{d}x} \tag{1.32}$$

and since $a \ge a_1 > 0$, the last term is ≥ 0. Therefore

$$F(w) \;\ge\; F(u) \qquad \forall\, w \in V. \tag{1.33}$$

Intuition: deviating from equilibrium always costs energy

Equation (1.32) is the whole content of the minimum potential energy principle, and it is worth reading slowly.

Take the true equilibrium shape u and perturb it by *any* admissible amount v. The potential energy does not change by an amount proportional to v — that first-order term is exactly (A)−(B), and it vanishes *because u is in equilibrium*. What is left is a purely second-order term, $\frac{1}{2}\int a(v')^2$, which is the strain energy stored by the perturbation itself, and it is always positive.

So: **the equilibrium configuration sits at the bottom of an energy well.** Any wrong shape you propose stores extra strain energy and has higher potential energy, by precisely the strain energy of your error. This is also the algebraic ancestor of Céa's lemma and of "FEM is too stiff": when the course later restricts to V_h, the best you can do is the shape in V_h that gets furthest down this well, and it can never get lower than u itself.

Recall: this is just completing the square

In finite dimensions with $\boldsymbol{K}$ symmetric positive definite, $F(\boldsymbol{w}) = \frac{1}{2}\boldsymbol{w}^T\boldsymbol{K}\boldsymbol{w} - \boldsymbol{w}^T\boldsymbol{F}$. Putting $\boldsymbol{w} = \boldsymbol{u} + \boldsymbol{v}$ with $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F}$ gives

$$F(\boldsymbol{u}+\boldsymbol{v}) = F(\boldsymbol{u}) + \boldsymbol{v}^T(\boldsymbol{K}\boldsymbol{u} - \boldsymbol{F}) + \tfrac{1}{2}\boldsymbol{v}^T\boldsymbol{K}\boldsymbol{v} = F(\boldsymbol{u}) + \tfrac{1}{2}\boldsymbol{v}^T\boldsymbol{K}\boldsymbol{v},$$

which is (1.32) term for term. Solving $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{F}$ and minimising F are the same task — exactly what Theorem 1.3.1 says in the continuous setting.

► **Unclear in my notes:** My notes conclude "$F(w) > F(u)$" with a strict inequality, but (1.32) gives equality when $v = 0$ (i.e. $w = u$). It should be $F(w) \geq F(u)$, with equality iff $w = u$. Strictness holds only for $w \neq u$.

② **(Min Prob) ⇒ (variational problem).** Assume $u \in V$ satisfies $F(u) \leq F(w)$ for all $w \in V$. Let

$$g(\varepsilon) = F(u + \varepsilon v), \qquad \text{where } v \in V \text{ is arbitrary but fixed.} \tag{1.34}$$

Expanding,

$$\begin{aligned} g(\varepsilon) &= \frac{1}{2}\int_0^1 a\,(u' + \varepsilon v')^2\,\mathrm{d}x - \int_0^1 f\,(u+\varepsilon v)\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 a(u')^2\,\mathrm{d}x + \varepsilon\int_0^1 au'v'\,\mathrm{d}x + \frac{1}{2}\varepsilon^2\int_0^1 a(v')^2\,\mathrm{d}x - \int_0^1 fu\,\mathrm{d}x - \varepsilon\int_0^1 fv\,\mathrm{d}x. \end{aligned} \tag{1.35}$$

$g(\varepsilon)$ is a quadratic polynomial in ε with a positive quadratic term. The unique global minimiser happens at $\varepsilon = 0$, so $g'(0) = 0$. Since

$$g'(\varepsilon) = \int_0^1 a\,u'v'\,\mathrm{d}x - \int_0^1 fv\,\mathrm{d}x + \varepsilon\int_0^1 a\,(v')^2\,\mathrm{d}x, \tag{1.36}$$

setting $\varepsilon = 0$ gives

$$g'(0) = \int_0^1 a\,u'v'\,\mathrm{d}x - \int_0^1 fv\,\mathrm{d}x = 0, \tag{1.37}$$

which is exactly the variational problem. Since v was arbitrary, the variational problem is completely equivalent to this energy minimisation. □

► **Unclear in my notes:** Page 19 writes $g'(\varepsilon) = \int u'v'\,\mathrm{d}x - \int fv\,\mathrm{d}x$, dropping the coefficient a from the first term, and the $+$ before the ε term is missing. Both are present on the following page, so I have restored them in (1.36).

Intuition: $g'(0) = 0$ is $\delta\Pi = 0$

This direction is the *first variation*, written out longhand. When a mechanics text writes $\delta\Pi = 0$, the operation it means is precisely

$$\delta\Pi = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\Pi(u + \varepsilon\,\delta u)\bigg|_{\varepsilon=0} = 0.$$

The trick of introducing a scalar ε, differentiating an ordinary one-variable function, and

setting $\varepsilon = 0$ is how a "variation" is made rigorous: it converts a question about a functional on an infinite-dimensional space into single-variable calculus along one line through u. v is again the virtual displacement, and "v arbitrary but fixed" is "for an arbitrary admissible virtual displacement." So this direction says: *minimum potential energy* $\Rightarrow$ *virtual work*. Direction ① said the converse. Together, they are the statement that the energy principle and the virtual-work principle are two descriptions of one fact.

Where this trips people up: the positive ε^2 coefficient is doing real work

Note the exact phrase: "a quadratic polynomial in ε *with a positive quadratic term*." That coefficient is $\frac{1}{2}\int a(v')^2$, and it is positive only because $a \geq a_1 > 0$. It is the same coercivity assumption that gave uniqueness in Lecture 2.

Why it matters: $g'(0) = 0$ alone only makes u a *stationary* point. A parabola opening upward makes it a minimum; if the quadratic coefficient could be zero or negative, $g'(0) = 0$ would still hold at a saddle or a maximum, and (1.30) would be false.

You have met the moment this fails. In buckling, the second variation of the potential energy loses positive-definiteness at the critical load — the equilibrium is still stationary, but it stops being a minimum, and the structure snaps to a different configuration. Everything in this course assumes we are safely on the positive-definite side of that.

Translation: the three formulations

Math department writes	**You should hear**
strong form $-(au')' = f$	pointwise equilibrium of a differential slice
variational problem $\int au'v' = \int fv \ \forall v$	principle of virtual work, $\delta W_{\text{int}} = \delta W_{\text{ext}}$
min problem $F(u) \leq F(w)$	principle of minimum total potential energy
$g'(0) = 0$	$\delta\Pi = 0$, stationary potential energy
ε^2 coefficient > 0	$\delta^2\Pi > 0$: the equilibrium is stable

Intuition: why this equivalence is worth a whole lecture

Three problems are now known to be the same object: the BVP (Lecture 2), the variational problem, and the minimisation problem. That is not bookkeeping — each gives a different way to *build* a numerical method:

- From the variational form, restrict v to $V_h \Rightarrow$ **Galerkin**.
- From the energy form, minimise F over $V_h \Rightarrow$ **Rayleigh–Ritz**.

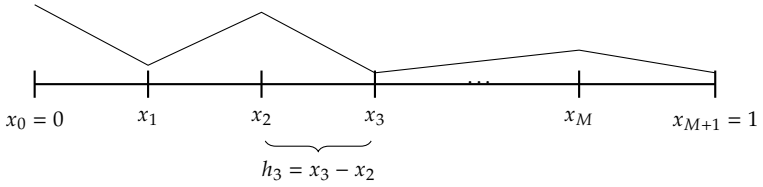
For this problem — symmetric a, positive definite — the two produce the *identical* linear system. That is why the mechanics literature calls FEM "Rayleigh–Ritz" and the mathematics literature calls it "Galerkin," and why both are right.

The equivalence breaks when the operator is not symmetric: an advection term bu' has no energy functional to minimise, so Rayleigh–Ritz has nothing to work with, while Galerkin still applies. That is the first hint of why the course will keep proving things about $a(\cdot,\cdot)$ rather than about F.

Our first finite element method

Idea: solve the variational problem (or the Min problem) approximately by replacing V with a finite-dimensional subspace V_h.

Let $\mathcal{T}_h$ be a mesh on $(0,1)$:



Let

$$h_j = x_j - x_{j-1}, \qquad j = 1, 2, \ldots, M+1. \tag{1.38}$$

The simplest $V_h \subset V$ is the **space of continuous piecewise linear polynomials on $\mathcal{T}_h$**. This is called the

CG(1) method Continuous Galerkin method with polynomials of degree 1. (1.39)

In mechanics terms: CG(1) is the 2-node bar element

Strip the name and this is the element you have built by hand. "Continuous piecewise linear on $\mathcal{T}_h$" means: straight-line displacement within each element, matching at the nodes. That is the linear 2-node bar — constant strain per element, and in 2D its cousin is the constant-strain triangle.
Two words carry weight:

- **Continuous**: the pieces join at the nodes. This is what makes $V_h \subset V$ ("conforming"), and Lecture 2 explained why — a jump would mean infinite strain energy, so a discontinuous V_h would not be a subspace of V at all. It is your C^0 compatibility requirement.
- **(1)**: polynomial degree. CG(2) is the 3-node quadratic bar, and so on. Raising it is p-refinement; shrinking h_j is h-refinement.

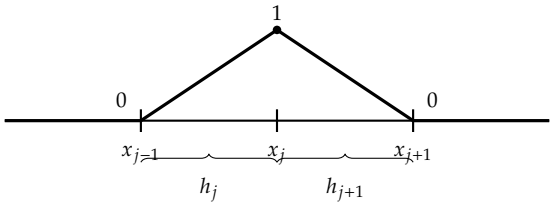
Note also that the mesh spacing h_j is allowed to vary — the elements need not be equal length. That is local mesh refinement, available from the very first definition.

Any function in V_h can be uniquely written as

$$U(x) = \sum_{j=1}^{M} \xi_j \, \varphi_j(x), \tag{1.40}$$

where the ξ_j are coefficients and $\varphi_j(x)$ is the hat function associated with x_j:

$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h_j}, & x_{j-1} \le x \le x_j, \\ \dfrac{x_{j+1} - x}{h_{j+1}}, & x_j \le x \le x_{j+1}, \\ 0, & \text{everything else, outside those limits.} \end{cases} \tag{1.41}$$



Intuition: the coefficients *are* the nodal displacements

The definition (1.41) is engineered so that

$$\varphi_j(x_i) = \delta_{ij} \qquad \text{(one at its own node, zero at every other node)},$$

and therefore, evaluating (1.40) at x_i,

$$U(x_i) = \sum_j \xi_j \, \varphi_j(x_i) = \xi_i.$$

So ξ_j is not an abstract expansion coefficient — it is **the displacement of node** j. This is the property that makes FEM usable: the unknowns you solve for are physically meaningful, boundary conditions are imposed by setting nodal values directly, and the answer can be plotted without post-processing. A basis without this property (a Fourier basis, say) gives coefficients that mean nothing individually.
This is called a *nodal* or *Lagrange* basis, and ξ is your DOF vector $\boldsymbol{U}$.

Where this trips people up: why the sum starts at $j = 1$ and stops at M

The mesh has nodes $x_0, \ldots, x_{M+1}$ — that is $M + 2$ nodes and $M + 1$ elements. But (1.40) sums only $j = 1, \ldots, M$: the *interior* nodes. The hats φ_0 and φ_{M+1} at the two endpoints are deliberately left out.
That omission *is* the boundary condition. Every function built from interior hats automatically satisfies $U(0) = U(1) = 0$, so $V_h \subset V$ by construction rather than by an extra constraint. In solver terms, this is eliminating the fixed DOFs from the system instead of penalising them, and it is why $\dim V_h = M$ and the stiffness matrix will be $M \times M$.
If the problem had $u(0) = g \neq 0$ you would put φ_0 back with its coefficient *fixed* at g and move its column to the right-hand side — which is the discrete version of the trial/test distinction flagged in Lecture 2.

Intuition: what happens next, and why the mesh makes K sparse

The lecture stops at the definition of V_h, but the next step is mechanical. Substituting $U = \sum_j \xi_j \varphi_j$ into (1.28) and testing against each φ_i turns the variational problem into

$$\boldsymbol{K\xi} = \boldsymbol{F}, \qquad K_{ij} = \int_0^1 a \, \varphi_j' \, \varphi_i' \, \mathrm{d}x, \qquad F_i = \int_0^1 f \, \varphi_i \, \mathrm{d}x.$$

Two features fall straight out of (1.41):

- φ_j is nonzero only on $[x_{j-1}, x_{j+1}]$, so $K_{ij} = 0$ unless $|i - j| \leq 1$: the stiffness matrix is **tridiagonal**. Sparsity is a consequence of choosing basis functions with small support, nothing more.
- φ_j' is *constant* on each element ($\pm 1/h$), so the element integrals can be done exactly by hand — which is why the classical bar-element stiffness $\frac{EA}{h}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ looks the way it does.

1.4 The discrete system: stiffness matrix and load vector

Lecture 4 — Monday, August 31, 2026 Math 6660 — OneNote export 4th.pdf

The big picture: Lecture 4

How does an integral identity become $\boldsymbol{KU} = \boldsymbol{F}$?

- Substitute the hat basis into the weak form. The bilinear form evaluated on two basis functions *is* the stiffness entry K_{ij}; the load functional *is* F_i. There is nothing else in assembly.
- **Galerkin orthogonality**: the error is orthogonal, in the energy inner product, to everything the mesh can represent. So the FEM answer is the *best* member of V_h, not just some member.
- For $a \equiv 1$ on a uniform mesh, $\boldsymbol{K}$ comes out as the classical central-difference stencil — same matrix, different reasoning.

Use it for: Reading assembly code, and knowing why refining a mesh can never make the energy-norm error worse.

Setup

2-pt BVP. $-(au')' = f$ on $(0,1)$, $u(0) = u(1) = 0$.

$$\Downarrow \text{ variational problem}$$

Find $u \in V$ such that

$$\int_0^1 a\, u'v'\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x \qquad \forall\, v \in V, \tag{1.42}$$

where $V = \left\{v : \int_0^1 v^2\,\mathrm{d}x < \infty,\ \int_0^1 (v')^2\,\mathrm{d}x < \infty,\ v(0) = v(1) = 0\right\}$.

Mesh $\mathcal{T}_h$. Grid points

$$x_0 = 0,\ x_1,\ x_2,\ \ldots,\ x_M,\ x_{M+1} = 1.$$

FEM space.

$$V_h := \left\{ v \in V \ :\ v|_{[x_i, x_{i+1}]} \in \text{polynomial of degree 1},\ \ \forall\, i = 0, \ldots, M \right\} \tag{1.43}$$

FEM. Find $U \in V_h \subset V$ such that

$$\int_0^1 a\, U'v'\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x \qquad \forall\, v \in V_h. \tag{1.44}$$

Intuition: the FEM problem is the variational problem with V swapped for V_h

Compare the two lines: identical, except $V \to V_h$. Nothing else changed — not the equation, not the bilinear form, not the load. **That single substitution is the entire finite element method.**
This is the second arrow of Lecture 3's chain, and it is where approximation finally enters. The first arrow (strong → weak) cost nothing.

A basis for V_h

$\varphi_1(x), \varphi_2(x), \ldots, \varphi_M(x)$, where

$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h_j}, & x_{j-1} < x < x_j, \\ \dfrac{x_{j+1} - x}{h_{j+1}}, & x_j < x < x_{j+1}, \end{cases} \tag{1.45}$$

$$V_h = \operatorname{span}\{\varphi_1, \varphi_2, \ldots, \varphi_M\}, \qquad \dim V_h = M. \tag{1.46}$$

If $U \in V_h$ then

$$U(x) = \sum_{j=1}^{M} \xi_j \, \varphi_j(x), \qquad \text{where} \qquad \xi_j = U(x_j), \quad j = 1, \ldots, M. \tag{1.47}$$

► **Unclear in my notes:** Page 2 writes the second branch of (1.45) as $(x_{j+1} - x_j)/h_{j+1}$. That numerator is just h_{j+1}, so the branch would be the constant 1 — not a hat. It must be $(x_{j+1} - x)/h_{j+1}$, which is what Lecture 3 had. Transcribed corrected.

In mechanics terms: ξ_j is the displacement of node j

$\xi_j = U(x_j)$ is stated outright here, and it is the property that makes the method usable: the unknowns you solve for *are* the nodal displacements, not abstract coefficients. It follows from $\varphi_j(x_i) = \delta_{ij}$.
$\dim V_h = M$ counts *interior* nodes only — φ_0 and φ_{M+1} are absent from the basis, which is how $U(0) = U(1) = 0$ is enforced. In solver terms: the fixed DOFs were eliminated, not penalised.

Galerkin orthogonality

$$\int_0^1 a\, u'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x, \qquad u, v \in V, \tag{1.48}$$

$$\int_0^1 a\, U'v' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x, \qquad U, v \in V_h. \tag{1.49}$$

Subtract. Since $V_h \subset V$, (1.48) also holds for $v \in V_h$:

$$\boxed{\int_0^1 a\,(u - U)'\, v' \,\mathrm{d}x = 0 \qquad \text{for all } v \in V_h} \tag{1.50}$$

Intuition: the error is invisible to the mesh

$a(\cdot,\cdot)$ is symmetric and positive definite, so it is an *inner product* — the energy inner product. Read (1.50) in that language:

$$\big\langle\, u - U,\, v \,\big\rangle_a = 0 \qquad \forall\, v \in V_h \qquad \Longleftrightarrow \qquad (u - U) \perp V_h.$$

So U is the **orthogonal projection** of u onto V_h — the closest point in V_h to u, measured in energy. Not merely a good approximation: *the best one the mesh can hold*.
Pythagoras then gives $\|u\|^2 = \|U\|^2 + \|u - U\|^2$, hence $\|U\| \le \|u\|$: the discrete solution stores *less* energy than the true one. **A conforming displacement-based FE model is too stiff**, and refining softens it monotonically toward the truth. That is the theorem behind the rule of thumb.

Note where $V_h \subset V$ was used: without it, (1.48) could not be tested against $v \in V_h$ and the subtraction would be illegal. Conformity is not decoration.

The linear system

Put (1.47) into (1.44) and take $v = \varphi_i$, $i = 1, \dots, M$:

$$\int_0^1 a\Big(\sum_{j=1}^{M} \xi_j \varphi_j\Big)' \varphi_i' \,\mathrm{d}x = \int_0^1 f\,\varphi_i \,\mathrm{d}x \quad \Longrightarrow \quad \sum_{j=1}^{M} \xi_j \int_0^1 a\,\varphi_i' \varphi_j' \,\mathrm{d}x = \int_0^1 f\,\varphi_i \,\mathrm{d}x. \tag{1.51}$$

This is an $M \times M$ linear system:

$$\boxed{\boldsymbol{K}\,\boldsymbol{\xi} = \boldsymbol{b}} \qquad K_{ij} = \int_0^1 a\,\varphi_i'\,\varphi_j' \,\mathrm{d}x, \qquad b_i = \int_0^1 f\,\varphi_i \,\mathrm{d}x, \tag{1.52}$$

with $\boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_M)^T$.

In mechanics terms: this is the stiffness matrix, literally

The lecturer calls $\boldsymbol{K}$ the **stiffness matrix** and lists three properties. Each is a fact you already use:

	lecture	you call it
(1)	symmetric: test and trial from the same space	$\boldsymbol{K} = \boldsymbol{K}^T$, Betti reciprocity
(2)	positive definite: $a > 0$	no rigid-body modes; $\boldsymbol{K}$ invertible
(3)	tridiagonal: compact support $[x_{i-1}, x_{i+1}]$	banded/sparse $\boldsymbol{K}$

(1) holds because $a(u, v) = a(v, u)$ — the same self-adjointness proved in Lecture 1. Break it (add advection) and $\boldsymbol{K}$ stops being symmetric. (3) is purely geometric: φ_i lives only on $[x_{i-1}, x_{i+1}]$, so

$$\int_0^1 a\,\varphi_i'\varphi_j' \,\mathrm{d}x \neq 0 \qquad \text{only for } j = i-1,\ i,\ i+1.$$

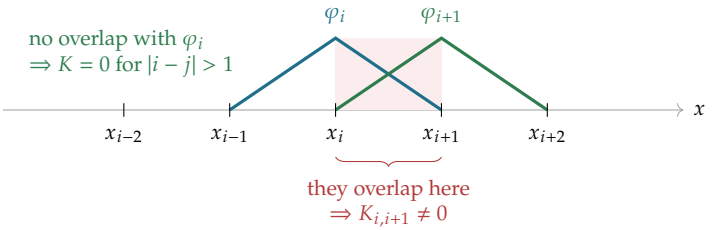


Figure 1.3: Adjacent hats overlap on one element, so only $j = i-1, i, i+1$ give a nonzero entry. Sparsity is a consequence of small support, nothing more.

Worked example: $a \equiv 1$, uniform mesh $h_j \equiv h$

Diagonal. $\varphi_i' = +1/h$ on (x_{i-1}, x_i) and $-1/h$ on (x_i, x_{i+1}), so $(\varphi_i')^2 = 1/h^2$ on both:

$$\begin{aligned} K_{ii} &= \int_0^1 1 \cdot \varphi_i'\varphi_i' \,\mathrm{d}x = \int_{x_{i-1}}^{x_i} (\varphi_i')^2 \,\mathrm{d}x + \int_{x_i}^{x_{i+1}} (\varphi_i')^2 \,\mathrm{d}x \\ &= \frac{1}{h^2}(x_i - x_{i-1}) + \frac{1}{h^2}(x_{i+1} - x_i) = \frac{1}{h} + \frac{1}{h} = \frac{2}{h}. \end{aligned} \tag{1.53}$$

Off-diagonal. On (x_i, x_{i+1}): $\varphi_i' = -1/h$ while $\varphi_{i+1}' = +1/h$, so the product is $-1/h^2$ over a length h:

$$K_{i,i+1} = \int_{x_i}^{x_{i+1}} \varphi_i' \, \varphi_{i+1}' \, \mathrm{d}x = -\frac{1}{h^2} \cdot h = -\frac{1}{h}. \tag{1.54}$$

Load vector. By the composite trapezoidal rule on $[x_{i-1}, x_{i+1}]$, using $\varphi_i(x_{i\pm1}) = 0$ and $\varphi_i(x_i) = 1$:

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f \, \varphi_i \, \mathrm{d}x \approx \frac{h}{2}\Big[f(x_{i-1}) \underbrace{\varphi_i(x_{i-1})}_{0} + 2 f(x_i) \underbrace{\varphi_i(x_i)}_{1} + f(x_{i+1}) \underbrace{\varphi_i(x_{i+1})}_{0} \Big] = h \, f(x_i). \tag{1.55}$$

Result.

$$\boldsymbol{K} = \frac{1}{h} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 \end{bmatrix}, \qquad \boldsymbol{b} = h\big(f(x_1), f(x_2), \ldots, f(x_M)\big)^T. \tag{1.56}$$

In mechanics terms: this is your assembled bar stiffness

Take the 2-node bar element of length h with $EA = a = 1$. Its element stiffness is the one you derive by hand:

$$\boldsymbol{k}^e = \frac{EA}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Assemble: node i is shared by elements $(i{-}1, i)$ and $(i, i{+}1)$, so it collects $\frac{1}{h} + \frac{1}{h} = \frac{2}{h}$ on the diagonal, and $-\frac{1}{h}$ from the single shared element off-diagonal. That is exactly (1.56).
So the lecturer's $\boldsymbol{K}$ and your assembled bar stiffness are the same matrix — derived from a variational principle here, from equilibrium of springs in a mechanics course. And $\boldsymbol{b} = h\,f(x_i)$ is the consistent load vector, which trapezoidal quadrature has turned into the *lumped* one.

Note: this is exactly the central finite difference method

FDM will provide the same matrix as FEM for the 2-pt BVP. Row i of (1.56) reads

$$\frac{1}{h}\big(-\xi_{i-1} + 2\xi_i - \xi_{i+1}\big) = h\, f(x_i) \qquad \Longleftrightarrow \qquad \frac{-\xi_{i-1} + 2\xi_i - \xi_{i+1}}{h^2} = f(x_i), \tag{1.57}$$

which is the $O(h^2)$ central difference approximation of $-u'' = f$.

► **Unclear in my notes:** Page 6 reads "FDM will also provide the same *Mathematic*..." — I think this is "the same *matrix*". Confirm.

Where this trips people up: same matrix here, different methods in general

The two coincide only because of a stack of special choices: $a \equiv 1$, a *uniform* mesh, linear elements, and trapezoidal quadrature for $\boldsymbol{b}$. Change any one and they part company:

change	consequence
non-uniform h_j	FEM handles it directly; FDM needs a new stencil
variable $a(x)$	FEM integrates a exactly; FDM must average it
exact quadrature for $\boldsymbol{b}$	$\boldsymbol{b} \neq h\,f(x_i)$ — the systems differ
2D/3D, unstructured mesh	FDM has no counterpart

And they were *derived* differently even here: FDM discretises the *equation*, FEM discretises the *space*. That distinction — the first sentence of Lecture 1 — is why FEM keeps working when the mesh stops being a grid. Agreement on this one problem is a coincidence worth noticing, not a general fact.

1.5 Nonhomogeneous Dirichlet and Neumann BCs

Lecture 5 — Wednesday, September 2, 2026 Math 6660 — 5th.pdf

The big picture: Lecture 5

What changes when the ends are not clamped at zero?

- **Prescribed displacement** (Dirichlet, "essential"): construct a function that already satisfies it and solve for the correction. This is the *lifting* trick, and in code it is the constrained-dof step.
- **Half-hat** basis functions appear at the boundary nodes.
- **Prescribed force** (Neumann, "natural"): does not constrain the space at all. It falls out of the integration by parts as an extra term on the right-hand side.

Use it for: The essential-vs-natural distinction in one sentence: displacement BCs modify $\boldsymbol{K}$ and the space; force BCs modify $\boldsymbol{F}$ only.

Nonhomogeneous Dirichlet BCs

$$-(au')' = f \ \text{ in } (0,1), \qquad u(0) = \alpha, \quad u(1) = \beta. \tag{1.58}$$

Variational problem. Integrating by parts,

$$\int_0^1 au'v' \,\mathrm{d}x - \Big[au'v\Big]_0^1 = \int_0^1 fv \,\mathrm{d}x. \tag{1.59}$$

To make the boundary term vanish, define two spaces:

$$V = \Big\{v : \textstyle\int_0^1 \big[v^2 + (v')^2\big] \,\mathrm{d}x < \infty, \ \ v(0) = \alpha, \ v(1) = \beta\Big\}, \tag{1.60}$$

$$V_0 = \Big\{v : \textstyle\int_0^1 \big[v^2 + (v')^2\big] \,\mathrm{d}x < \infty, \ \ v(0) = 0 = v(1)\Big\}. \tag{1.61}$$

$$\text{Find } u \in V \text{ such that } \int_0^1 au'v' \,\mathrm{d}x = \int_0^1 fv \,\mathrm{d}x \qquad \forall\, v \in V_0. \tag{1.62}$$

► **Unclear in my notes:** My notes write the integrability condition as $< \alpha$, but α is already the boundary value. It must be $< \infty$.

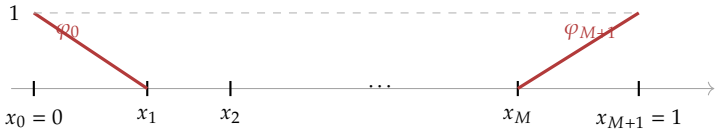
Where this trips people up: trial ≠ test — this is the case Lecture 2 warned about

$u \in V$ but $v \in V_0$. Lecture 2 noted they coincide only when the essential BC is homogeneous; (1.58) is exactly the case where they part. V is not even a vector space — u_1+u_2 has boundary value 2α.

In mechanics terms

V = displacement fields that satisfy the prescribed support movement. V_0 = virtual displacements, which must not violate the supports. A settling foundation moves the structure; the virtual motion still respects it.

Half-hat functions



Use the same basis $\varphi_j(x)$, $j = 1, \dots, M$, and introduce two *half-hat* functions:

$$\varphi_0(x) = \frac{x_1 - x}{x_1 - x_0}, \quad x \in (x_0, x_1); \qquad \varphi_{M+1}(x) = \frac{x - x_M}{x_{M+1} - x_M}, \quad x \in [x_M, x_{M+1}]. \tag{1.63}$$

Trial function.

$$U(x) = \alpha\, \varphi_0(x) + \beta\, \varphi_{M+1}(x) + \sum_{j=1}^{M} \xi_j\, \varphi_j(x). \tag{1.64}$$

Test functions. $v = \varphi_i(x)$, $i = 1, \dots, M$.

homogeneous means the boundary is zero: $\alpha = \beta \neq 0$ is *not* homogeneous; $\alpha = \beta = 0$ is homogeneous.

Intuition: the first two terms are a prescribed-displacement lift

$\alpha\varphi_0 + \beta\varphi_{M+1}$ is a fixed, known function that carries the boundary data — nothing in it is solved for. Only the ξ_j are unknowns, so $\dim V_h$ is still M.
This is what a solver does when you prescribe a nonzero DOF: it does not add an equation, it moves that column's contribution to the right-hand side.

The modified system

Substituting (1.64) into (1.62):

$$\sum_{j=1}^{M} \xi_j \int_0^1 a\, \varphi_i' \varphi_j'\, \mathrm{d}x \;+\; \alpha \int_0^1 a\, \varphi_0' \varphi_i'\, \mathrm{d}x \;+\; \beta \int_0^1 a\, \varphi_{M+1}' \varphi_i'\, \mathrm{d}x \;=\; \int_0^1 f\, \varphi_i\, \mathrm{d}x. \tag{1.65}$$

Moving the known terms across, and using $\varphi_0' = -1/h_1$, $\varphi_1' = +1/h_1$ on (x_0, x_1):

$$\boxed{\sum_{j=1}^{M} \xi_j \int_0^1 a\, \varphi_i' \varphi_j'\, \mathrm{d}x = \int_0^1 f\, \varphi_i\, \mathrm{d}x + \delta_{1i}\, \frac{\alpha}{h_1^2} \int_{x_0}^{x_1} a(x)\, \mathrm{d}x + \delta_{Mi}\, \frac{\beta}{h_{M+1}^2} \int_{x_M}^{x_{M+1}} a(x)\, \mathrm{d}x} \tag{1.66}$$

Intuition: the BC becomes a load on the first and last equation only

The Kronecker deltas confine the new terms to rows 1 and M. Every interior equation is untouched.
Reason: φ_0 overlaps only φ_1, so $\int a\, \varphi_0' \varphi_i' = 0$ for $i \geq 2$. Small support again.
Mechanically, $\boldsymbol{K}$ is unchanged and $\boldsymbol{F}$ picks up a reaction-like entry at the constrained nodes — exactly the "move the known DOF to the RHS" step.

Note. Dirichlet BCs are enforced *strongly* and modify the RHS ⇒ no error in the BCs. In FEM language, Dirichlet BCs are called **essential** BCs: they must be imposed in the definition of the function space V.

► **Unclear in my notes:** My notes read "No error in BC from BCs" — garbled. I have written "no error in the BCs," i.e. the boundary values are reproduced exactly.

Neumann BCs

$$-(au')' = f \ \text{ in } (0,1), \qquad \underbrace{u(0) = 0}_{\text{Dirichlet}}, \qquad \underbrace{a(1)u'(1) = g_1}_{\text{Neumann}}. \tag{1.67}$$

Here

$$V = \left\{ v : \int_0^1 \left(v^2 + (v')^2\right) \mathrm{d}x < \infty, \ \ v(0) = 0 \right\}. \tag{1.68}$$

Only $v(0) = 0$ is imposed — nothing is required at $x = 1$.

Integrating by parts:

$$\int_0^1 au'v' \,\mathrm{d}x - \underbrace{a(1)u'(1)}_{=g_1} v(1) + a(0)u'(0) \underbrace{v(0)}_{=0} = \int_0^1 fv \,\mathrm{d}x, \tag{1.69}$$

$$\Longrightarrow \quad \int_0^1 au'v' \,\mathrm{d}x - g_1 v(1) = \int_0^1 fv \,\mathrm{d}x \qquad \forall\, v \in V. \tag{1.70}$$

Intuition: the Neumann datum lands in the load vector

Compare with the Dirichlet case. There the boundary term was *killed* by forcing $v = 0$; here it is *kept*, because g_1 is known. It moves to the RHS as $+g_1 v(1)$ — a load, not a constraint. $g_1 = a(1)u'(1) = N(1)$ is the end force. So "apply a traction" really is just adding a term to $\boldsymbol{F}$, and that is why you constrain displacement DOFs explicitly in a solver but merely *apply* a traction.

Recovering the BVP (Neumann case)

Q: is the solution of the variational problem also a solution of the 2-pt BVP?

Start from (1.70) and reverse the integration by parts:

$$-\int_0^1 (au')'v \,\mathrm{d}x + a(1)u'(1)v(1) = \int_0^1 fv \,\mathrm{d}x + g_1 v(1). \tag{1.71}$$

This must hold for all $v \in V$, *including* all $v \in V$ with $v(1) = 0$. For those,

$$\int_0^1 fv \,\mathrm{d}x = -\int_0^1 (au')'v \,\mathrm{d}x \quad \forall\, v \in V \text{ with } v(1) = 0 \quad \Longrightarrow \quad -(au')' = f \ \text{ in } (0,1). \tag{1.72}$$

Putting that back leaves $a(1)u'(1)v(1) = g_1 v(1)$, and since $v(1)$ is arbitrary,

$$\boxed{a(1)u'(1) = g_1} \tag{1.73}$$

Intuition: the BC was never imposed — it fell out

Two-stage argument: restricting to $v(1) = 0$ recovers the PDE (Lecture 2's localisation), then releasing $v(1)$ recovers the Neumann condition.

g_1 was never put into V, yet the solution satisfies it. That is what **natural** means: satisfied automatically, not enforced. It is also why Neumann data is only met approximately by

the discrete solution — see the last note below.

CG(1) for the Neumann problem

$$V_h = \{v \in V : v|_{[x_j, x_{j+1}]} = \text{polynomial of degree } 1\} \tag{1.74}$$

$$V_h = \text{span}\{\varphi_1, \varphi_2, \ldots, \varphi_M, \underbrace{\varphi_{M+1}}_{\text{new}}\} \tag{1.75}$$

φ_{M+1} is the half-hat at the right end point. Trial function:

$$U(x) = \sum_{j=1}^{M+1} \xi_j \varphi_j(x), \qquad \int_0^1 aU'v'\,dx = \int_0^1 fv\,dx + g_1 v(1) \quad \forall\, v \in V_h, \tag{1.76}$$

with $v = \varphi_i$, $i = 1, 2, \ldots, M, \underbrace{M+1}_{\text{new}}$. Note

$$\varphi_i(1) = 0 \;\; \forall\, i = 1, \ldots, M, \qquad \varphi_{M+1}(1) = 1 \;\; \text{— the only one nonzero at } x = 1. \tag{1.77}$$

Intuition: $M+1$ unknowns now, not M

x_{M+1} is no longer prescribed, so its nodal value is an unknown and the system grows by one. Dirichlet removes a DOF; Neumann does not.
Since φ_{M+1} is the only basis function nonzero at $x = 1$, the term $g_1 v(1)$ contributes to exactly one entry of $\boldsymbol{F}$ — the last.

At $x = 1$, what is the error? $\Rightarrow$ Not zero.

Where this trips people up: Dirichlet is exact at the boundary, Neumann is not

Strongly enforced (Dirichlet): the boundary value is built into the space, so the discrete solution matches it *exactly*. Weakly enforced (Neumann): the condition only holds in the limit, so $a(1)U'(1) \neq g_1$ on a finite mesh.
You have seen this: prescribed displacements come back exactly, while reaction forces and surface tractions carry discretisation error and need a finer mesh.

1.6 Neumann assembly and the first error estimate

Lecture 6 — Friday, September 4, 2026 Math 6660 — `6th.pdf`

The big picture: Lecture 6

How wrong is the FEM answer, and measured how?

- Assemble the Neumann problem explicitly first — the boundary rows are where the natural BC shows up.
- Then error analysis begins. Two **norms** matter: the energy norm (strain energy stored in the error) and the L^2 norm (plain size of the error).
- First **a priori** estimate: the FEM error is bounded by the *interpolation* error — by how well the mesh could represent u at best, regardless of the method.

Use it for: Convergence studies: knowing what slope to expect before you run one.

Setup, restated

$$-(au')' = f \ \text{ in } (0,1), \qquad \underbrace{u(0) = 0,}_{\text{Dirichlet}} \qquad \underbrace{a(1)u' = g_1}_{\text{Neumann}} \tag{1.78}$$

$$V = \Big\{ v : \int \big[v^2 + (v')^2\big]\,\mathrm{d}x < \infty, \qquad \underbrace{v(0) = 0}_{\text{strongly enforced}} \quad \Big\} \tag{1.79}$$

CG(1) method.

$$U(x) = \sum_{j=1}^{M+1} \xi_j \varphi_j(x) \qquad \text{trial solution; the +1 is because the last node is unknown} \tag{1.80}$$

$$\int_0^1 aU'v'\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x + \underbrace{g_1 v(1)}_{\text{Neumann — weakly enforced}}, \qquad v = \varphi_i, \ \ i = 1, 2, \ldots, M+1. \tag{1.81}$$

Example: $a = 1$, uniform mesh $h_i \equiv h$

$$\sum_{j=1}^{M+1} \xi_j \int_0^1 \varphi_i' \varphi_j'\,\mathrm{d}x = \int_0^1 f\varphi_i\,\mathrm{d}x + g_1\varphi_i(1) \tag{1.82}$$

The new entries, all on the last row and column:

$$A_{(M,M+1)} = \int_{x_M}^{x_{M+1}} \varphi_{M+1}'\,\varphi_M'\,\mathrm{d}x = -\frac{1}{h^2}\int_{x_M}^{x_{M+1}} \mathrm{d}x = -\frac{1}{h}, \tag{1.83}$$

$$A_{(M+1,M)} = -\frac{1}{h} \qquad \Longrightarrow \qquad \text{symmetric matrix}, \tag{1.84}$$

$$A_{(M+1,M+1)} = \int_{x_M}^{x_{M+1}} \big(\varphi_{M+1}'\big)^2\,\mathrm{d}x = \frac{1}{h}. \tag{1.85}$$

So $\underline{\underline{A}}\,\underline{\xi} = \underline{b}$ with $\underline{\xi} = (\xi_1, \xi_2, \xi_3, \ldots, \xi_{M+1})$ and

$$A = \frac{1}{h}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & \boxed{1} \end{bmatrix}, \qquad A \in \mathbb{R}^{(M+1)\times(M+1)}, \ \text{symmetric, positive definite.} \tag{1.86}$$

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f\varphi_i\,\mathrm{d}x \ \ (i = 1, \ldots, M), \qquad b_{M+1} = \int_{x_M}^{x_{M+1}} f\varphi_{M+1}\,\mathrm{d}x + g_1. \tag{1.87}$$

Intuition: the last diagonal entry is 1, not 2

Every interior node has two elements attached, contributing $1/h$ each: hence $2/h$. The end node x_{M+1} has only *one* element, so it gets $1/h$.
The 2, −1 pattern was never a rule — it is just "sum of the elements meeting at this node." This is exactly element-by-element assembly: each element contributes $\frac{1}{h}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ to its two nodes, and the boundary node collects one copy instead of two.
Note also g_1 enters b_{M+1} alone, as a plain *nodal force* — the end traction assembled to the end node. No mystery.

Note: for the 2-pt BVP.

① Dirichlet BCs are **strongly** enforced (essential).

② Neumann or mixed BCs are **weakly** enforced (natural BCs).

Translation: the two BC types

Math department writes	You should hear
essential / Dirichlet / strong	prescribed displacement; built into V; removes a DOF; exact
natural / Neumann / weak	applied traction; appears in $\boldsymbol{F}$; adds no constraint; approximate

Error analysis

Two kinds of error estimates:

① **a priori**: error estimate in terms of derivatives of the exact solution, e.g.

$$\underbrace{\|u\|}_{} \ \ldots \ \| \underbrace{u}_{\text{exact}} - \underbrace{U}_{\text{FEA}} \| \leq \underbrace{C}_{\text{constant}} \overbrace{h}^{p}{}^{\text{order of shape function}} \|u^{(\ell)}\| \qquad \ell^{\text{th}} \text{ derivative of } u.$$

② **a posteriori**: "error estimates in terms of a residual", e.g. $\mathcal{L}(U) - f$. Useful for calculating error *during* the simulation ⇒ adaptive error control (adaptive mesh refinement).

Intuition: before vs. after

A priori uses u, which you do not have — it predicts the *rate* (h^p), so it tells you what refinement buys before you run anything. A posteriori uses U, which you do have — it says *where* the error is, which is what drives adaptive remeshing.

Norms

Let $e(x) := u(x) - U(x)$ be the error.

$$\text{"Energy" norm:} \qquad \|v\|_E = \sqrt{\int_0^1 a(x)\big[v'(x)\big]^2 \,\mathrm{d}x} \tag{1.88}$$

$$\text{weighted } L_2\text{-norm:} \qquad \|w\|_a = \sqrt{\int_0^1 a(x)\big[w(x)\big]^2 \,\mathrm{d}x} \tag{1.89}$$

$$\textbf{Note: } \|v\|_E = \|v'\|_a. \tag{1.90}$$

In mechanics terms

$\|v\|_E^2 = \int a(v')^2 = 2\times$ strain energy. Measuring error in the energy norm means measuring the strain energy of the error — i.e. how much stored energy your mesh is missing. It weights *strain* error, not displacement error, which is why stress converges more slowly than displacement.

Cauchy inequalities.

$$\left|\int_0^1 av'w'\,\mathrm{d}x\right| \le \|v'\|_a\|w'\|_a = \|v\|_E\|w\|_E, \tag{1.91}$$

$$\left|\int_0^1 vw\,\mathrm{d}x\right| \le \|v\|_a\|w\|_{a^{-1}}. \tag{1.92}$$

An a priori error estimate

CG(1) for homogeneous Dirichlet BCs: $-(au')' = f$ in $(0,1)$, $u(0) = u(1) = 0$.

Error calculation.

$$\int_0^1 aU'v'\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x \qquad \forall\, v \in V_h \subset V, \tag{1.93}$$

$$\int_0^1 au'v'\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x \qquad \forall\, v \in V. \tag{1.94}$$

Now subtract:

$$\boxed{\int_0^1 a\,(u-U)'\,v'\,\mathrm{d}x = 0 \qquad \forall\, v \in V_h} \qquad \textbf{Galerkin orthogonality} \tag{1.95}$$

Energy norm error estimate. Using (1.90),

$$\|u-U\|_E^2 = \|(u-U)'\|_a^2 = \int_0^1 a\,(u-U)'(u-U)'\,\mathrm{d}x. \tag{1.96}$$

For *any* $v \in V_h$, insert $-v+v$:

$$\begin{aligned}\|u-U\|_E^2 &= \int_0^1 a\,(u-U)'\,(u-v+v-U)'\,\mathrm{d}x \\ &= \int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x + \underbrace{\int_0^1 a\,(u-U)'(v-U)'\,\mathrm{d}x}_{=\,0 \text{ by Galerkin orthogonality}}.\end{aligned} \tag{1.97}$$

Note. $u - v \in V$, but $v - U \in V_h$ — which is what lets (1.95) kill the second term.

Where this trips people up: the trick is choosing which difference lies in V_h

$u-U$ is *not* in V_h (u is not a piecewise linear). But $v-U$ *is*, since both v and U are. Galerkin orthogonality only annihilates against members of V_h, so the split $u-U = (u-v)+(v-U)$ is engineered to put a V_h member in the second slot.

So, for any $v \in V_h$,

$$\|u-U\|_E^2 = \int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x. \tag{1.98}$$

Using Cauchy (1.91),

$$\int_0^1 a\,(u-U)'(u-v)'\,\mathrm{d}x \le \|(u-U)'\|_a\,\|(u-v)'\|_a = \|u-U\|_E\,\|u-v\|_E, \tag{1.99}$$

so $\|u-U\|_E^2 \le \|u-U\|_E\|u-v\|_E$, and cancelling one factor:

$$\boxed{\|u-U\|_E \;\le\; \|u-v\|_E \qquad \forall\, v \in V_h} \tag{1.100}$$

The FE solution U is the best solution, meaning *smallest in the energy norm*.

► **Unclear in my notes:** The middle line of my notes mixes the two norms ($\|(u-U)'\|_E$ where it should be $\|u-U\|_E$, since $\|v\|_E = \|v'\|_a$ already carries the derivative). Restored above.

Intuition: Céa's lemma with constant 1

Lecture 1's sample stated Céa with a factor M/α. Here it is *exactly* 1 — because the energy norm is built from the same a as the bilinear form, so continuity and coercivity constants are both 1 by construction.

Read (1.100) directly: U is the closest point of V_h to u in energy. No other piecewise linear on this mesh stores a more accurate energy. Combined with Galerkin orthogonality, U is the orthogonal projection of u onto V_h, and Pythagoras gives $\|U\|_E \le \|u\|_E$ — **the FE model is too stiff**, as always.

Practical reading: the solver is not the limit. Accuracy is set entirely by how well piecewise linears can represent u on this mesh. Better answers come from a better mesh, not a better solve.

1.7 A priori error estimates: energy norm and L^2

Lecture 7 — Wednesday, September 9, 2026 Math 6660 — 7th.pdf

The big picture: Lecture 7

What convergence rate, exactly?

- **Energy norm:** $O(h)$ for CG(1), and the constant is driven by u'' — curvature of the true solution. Flat solution, small error.
- **L^2 norm:** $O(h^2)$, one order better. The extra order is not free — it comes from a duality argument, not from the same estimate.
- The working tool throughout is the **interpolant** $\Pi_h u$: compare the FEM answer to it, never directly to u.

Use it for: Reading the slope on a log–log convergence plot and telling whether your element is behaving or you have a bug.

Setup

$$-(au')' = f \;\text{ in } (0,1), \qquad u(0) = u(1) = 0, \tag{1.101}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2\right] \mathrm{d}x < \infty,\; v(0) = v(1) = 0 \right\}, \qquad V_h \;\text{ for CG(1).} \tag{1.102}$$

(1) Energy norm

$$\|e\|_E^2 = \int_0^1 a\left[(u-U)'\right]^2 \mathrm{d}x. \tag{1.103}$$

Last time it was shown that the FEM solution satisfies

$$\|u-U\|_E \;\le\; \|u-v\|_E \qquad \forall\, v \in V_h \quad \text{(any other function in } V_h\text{)}. \tag{1.104}$$

Now take the test function

$$v = \Pi_h u \in V_h, \qquad \Pi_h u = \sum_{j=1}^{M} u(x_j)\,\varphi_j(x), \tag{1.105}$$

the projection of $u \in V$ into the subspace $V_h \subset V$: interpolate $u(x)$ onto the mesh using piecewise linear (continuous) functions.

► **Unclear in my notes:** Page 2 writes the sum as $\sum_{j=1}^{M} u(x_i)\varphi_i(x)$ — the index under the sum is j but the terms carry i. Transcribed with j throughout. The word before "functions" is hard to read; from context it must be *linear*, not *Green*.

Intuition: what $\Pi_h u$ is, and why it is the right test function

$\Pi_h u$ is the piecewise-linear function that *agrees with u at every node* — connect the dots of the exact solution with straight lines. It is computable in principle and, crucially, it is a legal member of V_h.

Why feed it into (1.104): that inequality holds for *every* $v \in V_h$, so it holds for the one we can estimate. Céa says U is at least as good as $\Pi_h u$, so bounding how badly a straight line approximates u on one element bounds the entire FE error. The solver never enters.

Note.

$$\|u - \Pi_h u\|_E = \|(u - \Pi_h u)'\|_a \le C\,\|h\,u''\|_a, \tag{1.106}$$

which just comes from approximating u' by a piecewise constant function; C is a constant that depends only on $a(x)$.

Theorem 1.7.1. *The FEM approximation $U \in V_h$ satisfies $\|u - U\|_E \le \|u - v\|_E$ for all $v \in V_h \subset V$. In particular there exists a constant C, depending only on $a(x)$, such that*

$$\boxed{\|u - U\|_E \le C\,\|h\,u''\|_a} \tag{1.107}$$

Note. This shows that CG(1) is *first order accurate in the energy norm*, provided $u'' \in L^2(0,1)$.

In mechanics terms: what first order buys you

Halve the mesh, halve the energy-norm error. And the energy norm weights *strain*, so (1.107) is a statement about stress accuracy.

Note the proviso: $u'' \in L^2$. At a re-entrant corner or a point load u'' is not square integrable, the right-hand side is infinite, and the estimate says nothing — which is exactly the case where uniform refinement stalls.

(2) $L^2(0,1)$ norm

For the same problem (1.101), consider the following **dual problem**:

$$-(a\phi')' = e, \qquad \phi(0) = \phi(1) = 0, \qquad \text{where } e = u - U. \tag{1.108}$$

Note. Because the original 2-pt BVP is self-adjoint, the primal and dual operators are the same.

Intuition: the dual problem, and why self-adjointness matters

The dual problem is an auxiliary BVP whose *load is the error itself*. Solve it and you obtain a ϕ engineered so that pairing it against e reproduces $\|e\|^2$ — which is the whole trick below. Self-adjointness (Lecture 1's Green's formula) is what makes the dual operator identical to the primal one. For a non-self-adjoint problem you would have to solve a genuinely

different equation here.

L^2-norm. Multiply (1.108) by e and integrate:

$$\|e\|^2 = \int_0^1 e^2 \,\mathrm{d}x = -\int_0^1 e\,(a\phi')'\,\mathrm{d}x$$
$$= \int_0^1 a\,e'\phi'\,\mathrm{d}x - \Big[a\,e\,\phi'\Big]_0^1, \tag{1.109}$$

and the boundary term vanishes because $e(0) = e(1) = 0$, so

$$\|e\|^2 = \int_0^1 a\,e'\phi'\,\mathrm{d}x. \tag{1.110}$$

Recall Galerkin orthogonality: $\int_0^1 a\,e'v'\,\mathrm{d}x = 0$ for all $v \in V_h$. Take $v = \Pi_h\phi$:

$$\|e\|^2 = \int_0^1 a\,e'\phi'\,\mathrm{d}x - \int_0^1 a\,e'\,(\Pi_h\phi)'\,\mathrm{d}x = \int_0^1 a\,e'\,(\phi - \Pi_h\phi)'\,\mathrm{d}x. \tag{1.111}$$

Intuition: subtracting zero is the whole move

$\int a\,e'(\Pi_h\phi)'\,\mathrm{d}x$ is zero by Galerkin orthogonality, so (1.111) subtracts nothing. But it converts ϕ' into the *interpolation error* $(\phi - \Pi_h\phi)'$, which is small — order h — whereas ϕ' itself is not small at all. That single substitution is what buys the extra power of h.

Cauchy's inequality.

$$\|e\|^2 \le \|e'\|_a\,\|(\phi - \Pi_h\phi)'\|_a \tag{1.112}$$
$$\|e\|^2 \le C_I\,h_{\max}\,\|e'\|_a \qquad \underbrace{\|\phi''\|_a}_{\text{interpolation error}} \tag{1.113}$$

But recall, from part (1),

$$\|e'\|_a = \|e\|_E \le C_I\,h_{\max}\,\|u''\|_a, \tag{1.114}$$

so

$$\|e\|^2 \le C_I^2\,h_{\max}^2\,\|u''\|_a\,\|\phi''\|_a = C_I^2\,h_{\max}^2\,\|u''\|_a\,\frac{\|\phi''\|_a}{\|e\|}\,\|e\|, \tag{1.115}$$

and dividing by $\|e\|$,

$$\|e\| \le C_I^2\,h_{\max}^2\,\|u''\|_a\,\frac{\|\phi''\|_a}{\|e\|}. \tag{1.116}$$

Let

$$S_a = \max_{\xi\neq 0}\frac{\|\psi''\|_a}{\|\xi\|}, \qquad \text{where} \qquad -(a\psi')' = \xi, \quad \psi(0) = \psi(1) = 0. \tag{1.117}$$

Note. If $a \equiv$ constant, then $-(a\psi')' = -a\psi'' = \xi$, so

$$\frac{\psi''}{\|\xi\|} = \frac{1}{a}. \tag{1.118}$$

► **Unclear in my notes:** Two things to check here. (i) The last line divides a *function* ψ'' by a *number* $\|\xi\|$; it should presumably read $\|\psi''\|/\|\xi\| = 1/a$, and the sign from $\psi'' = -\xi/a$ has been dropped. (ii) That $1/a$ comes out only if the numerator is the plain L^2 norm, not the a-weighted $\|\cdot\|_a$ used in (1.117) — with the weighted norm one gets $1/\sqrt{a}$. Ask which norm is meant in S_a.

Intuition: the payoff: L^2 converges one order faster

Combining (1.116) with (1.117) gives $\|e\| \le C_I^2 h_{\max}^2 S_a \|u''\|_a$ — **second** order in h, against first order for the energy norm (1.107).
Where the extra power came from: interpolation error was used *twice* — once on e through (1.114), once on ϕ through (1.111). Each contributes one factor of h. This is the duality (Aubin–Nitsche) argument.
Engineering reading: **displacements converge faster than stresses**. The energy norm measures strain error and gives $O(h)$; the L^2 norm measures displacement error and gives $O(h^2)$. That is why a displacement plot can look converged while the stress plot is still visibly wrong.

► **Unclear in my notes:** The notes stop at (1.118) without writing the final L^2 estimate explicitly. Confirm the intended statement is $\|e\| \le C_I^2 h_{\max}^2 S_a \|u''\|_a$.

1.8 The stability constant, and a posteriori error

Lecture 8 — Friday, September 11, 2026 Math 6660 — 8th.pdf

The big picture: Lecture 8

Why does the L^2 estimate get an extra power of h, and what is the difference between the two kinds of error bound?

- The duality argument needs a **stability constant** C_S: the auxiliary (dual) problem's solution must not blow up. That is what is proved here.
- The **Poincaré inequality** is the tool that delivers it — a function pinned at the boundary cannot be large if its derivative is small.
- Result: $\|e\| \le C\, h^2 \|u''\|$.
- Then the switch: **a priori** bounds involve u, which you do not have. **A posteriori** bounds involve U, which you just computed.

Use it for: Separating "what rate should this element give" from "how wrong is *this particular* run".

Recap

For $-(au')' = f$ on $(0,1)$, $u(0) = u(1) = 0$, with CG(1):

(1) Energy norm. $\|u-U\|_E^2 = \int_0^1 a\left[(u-U)'\right]^2 \mathrm{d}x$, and $\|u-U\|_E \le \|u-v\|_E$ for all $v \in V_h$. Taking $v = \Pi_h u$,

$$\|e\|_E \le C\, \|h\, u''\|_a. \tag{1.119}$$

(2) L^2 norm. $\|u-U\|^2 = \int_0^1 (u-U)^2 \,\mathrm{d}x$, and the summary bound from last lecture,

$$\|e\| \le C_I^2\, h_{\max}\, \|u''\|_a \, \frac{\|\phi''\|_a}{\|e\|}, \tag{1.120}$$

which still contains the unknown ϕ from the **dual problem**

$$-(a\phi')' = e, \qquad \phi(0) = \phi(1) = 0. \tag{1.121}$$

Intuition: what this lecture is for

(1.120) is not yet an error estimate — the right-hand side still mentions ϕ, which depends on the error you are trying to bound. The whole lecture is one job: show that $\|\phi''\|_a/\|e\|$ is bounded by a constant depending only on a, so it can be absorbed and the estimate closed.

Claim (the stability constant)

Theorem 1.8.1. *For*

$$-(a\psi')' = \xi, \qquad \psi(0) = \psi(1) = 0, \tag{1.122}$$

we have

$$\max_{\xi \neq 0} \frac{\|\psi''\|_a}{\|\xi\|} \;\leq\; C_S, \tag{1.123}$$

where C_S depends only on a.

Proof. Expanding (1.122), $-a\psi'' = a'\psi' + \xi$, so

$$\|a\psi''\| \leq \|\xi\| + \|a'\psi'\| \leq \|\xi\| + \|a'\|_\infty \|\psi'\|, \qquad \|a'\|_\infty = \max_{x\in[0,1]} |a'(x)|. \tag{1.124}$$

Take the BVP, multiply by ψ and integrate by parts:

$$-\int_0^1 \psi\,(a\psi')'\,\mathrm{d}x = \int_0^1 \xi\,\psi\,\mathrm{d}x \quad\Longrightarrow\quad \int_0^1 a\,(\psi')^2\,\mathrm{d}x = \int_0^1 \xi\,\psi\,\mathrm{d}x, \tag{1.125}$$

and since $a \geq a_{\min}$,

$$a_{\min}\int_0^1 (\psi')^2\,\mathrm{d}x \leq \int_0^1 \xi\,\psi\,\mathrm{d}x \quad\Longrightarrow\quad a_{\min}\|\psi'\|^2 \leq \|\xi\|\,\|\psi\|. \tag{1.126}$$

□

Aside: Poincaré inequality

Let $v \in V$. Then

$$v(x) = \int_0^x v'(y)\,\mathrm{d}y, \qquad |v| \leq \int_0^1 |v'|\,\mathrm{d}y \leq \sqrt{\int_0^1 (v')^2\,\mathrm{d}y}, \tag{1.127}$$

so $\int_0^1 v^2\,\mathrm{d}x \leq \int_0^1 (v')^2\,\mathrm{d}y$, i.e.

$$\|v\| \leq \|v'\|. \tag{1.128}$$

In mechanics terms: Poincaré is "pinned ends cannot move for free"

$v(0) = 0$ and v is built up by integrating its own slope, so a function that is small in slope cannot be large in value — it has nowhere to start from. In mechanics: constrain a bar at one end, and its displacement is controlled by its strain. Remove the constraint and (1.128) fails immediately: a rigid translation has $v' \equiv 0$ but $\|v\| \neq 0$.

Back to the proof

From (1.126) and Poincaré (1.128),

$$a_{\min}\|\psi'\|^2 \leq \|\xi\|\,\|\psi\| \leq \|\xi\|\,\|\psi'\| \qquad\Longrightarrow\qquad \|\psi'\| \leq \frac{\|\xi\|}{a_{\min}}. \tag{1.129}$$

Substituting into (1.124),

$$\begin{aligned} \|a\psi''\| &\le \|\xi\| + \|a'\|_\infty \frac{\|\xi\|}{a_{\min}} \\ &\le \left(1 + \frac{\|a'\|_\infty}{a_{\min}}\right) \|\xi\|. \end{aligned} \tag{1.130}$$

But $\sqrt{a_{\min}}\,\|\psi''\|_a \le \|a\psi''\|$, since

$$\|\psi''\|_a = \sqrt{\int_0^1 a\,(\psi'')^2\,\mathrm{d}x} \qquad \text{while} \qquad \|a\psi''\| = \sqrt{\int_0^1 a^2(\psi'')^2\,\mathrm{d}x}. \tag{1.131}$$

Therefore

$$\|\psi''\|_a \le \underbrace{\frac{1 + \dfrac{\|a'\|_\infty}{a_{\min}}}{\sqrt{a_{\min}}}}_{C_S} \|\xi\|, \qquad \text{so} \qquad \frac{\|\psi''\|_a}{\|\xi\|} \le C_S. \tag{1.132}$$

Intuition: C_S is a conditioning number

C_S grows when $a_{\min}$ is small (a nearly-zero-stiffness region) or when $\|a'\|_\infty$ is large (a sharply varying material). Both are the cases an engineer already treats as ill-conditioned — and the estimate degrades exactly there, not by coincidence.

Conclusion of the L^2 estimate

Feeding (1.132) into (1.120),

$$\boxed{\|e\| \le \underbrace{C_I^2 C_S}_{\text{constant}} \|h^2 u''\|_a} \qquad \text{or} \qquad \|e\| \le C_I^2\, C_S\, h_{\max}^2\, \|u''\|_a. \tag{1.133}$$

Assuming $\|u''\|_a$ is bounded, then $\|e\| = O(h^2)$.

Translation: the two a priori rates

Math department writes	You should hear
$\|e\|_E = O(h)$	strain/stress error: first order
$\|e\| = O(h^2)$	displacement error: second order
the extra power	interpolation used twice — on e and on ϕ

A posteriori error

(1) Energy norm.

$$\begin{aligned} \|e\|_E^2 = \|e'\|_a^2 &= \int_0^1 a\,(e')^2\,\mathrm{d}x \\ &= \int_0^1 a\,u'e'\,\mathrm{d}x - \int_0^1 a\,U'e'\,\mathrm{d}x \\ &= \int_0^1 f\,e\,\mathrm{d}x - \int_0^1 a\,U'e'\,\mathrm{d}x. \end{aligned} \tag{1.134}$$

Subtract from this the discrete equation, which vanishes for any $v \in V_h$:

$$\int_0^1 f\, v\, \mathrm{d}x - \int_0^1 a\, U' v'\, \mathrm{d}x = 0, \qquad \text{with } v = \Pi_h e \in V_h, \tag{1.135}$$

giving

$$\|e\|_E^2 = \int_0^1 f\,\big(e - \Pi_h e\big)\,\mathrm{d}x - \int_0^1 a\, U'\big(e - \Pi_h e\big)'\,\mathrm{d}x. \tag{1.136}$$

Intuition: why this one is computable

Everything on the right of (1.136) involves f and U — data and the computed solution. The exact u has disappeared. That is the whole point of an *a posteriori* bound: it can be evaluated after the solve, on the mesh you actually used, which is what makes adaptive refinement possible. Lecture 9 turns it into a per-element number.

1.9 A posteriori error and adaptive refinement

Lecture 9 — Monday, September 14, 2026 Math 6660 — `9th.pdf`

The big picture: Lecture 9

How does a solver decide where *to refine?*

- The **residual** $R(U) = f + (aU')'$ measures how badly the computed U fails the strong equation inside each element. It does *not* shrink under refinement — it stays $O(1)$.
- What shrinks is its weighted contribution: the element indicator E_j^2 multiplies the residual by the element size squared.
- **Equidistribution**: split the global error budget evenly over the elements and refine anything carrying more than its share. Repeat.

Use it for: This is literally the adaptive mesh refinement loop in a commercial solver, and E_j is the per-element indicator it colours on the screen.

Error analysis: where we are

a priori	① energy norm	② L^2 norm
a posteriori	① energy norm	

A posteriori error, energy norm

$$\|e\|_E^2 = \int_0^1 a\, e' e'\, \mathrm{d}x, \tag{1.137}$$

and from the FEM (Lecture 8),

$$\|e\|_E^2 = \int_0^1 f\,\big(e - \Pi_h e\big)\,\mathrm{d}x - \int_0^1 a\, U'\big(e - \Pi_h e\big)'\,\mathrm{d}x. \tag{1.138}$$

But from interpolation,

$$\underbrace{\|h^{-1}\big(e - \Pi_h e\big)\|_a}_{O(h^2)} \;\le\; C\; \underbrace{\|e'\|_a}_{O(h)}. \tag{1.139}$$

Note. Roughly speaking $R(U) = O(1)$.

Intuition: why the residual does not shrink

$R(U)$ is the amount by which the computed U fails to satisfy the strong equation pointwise. It stays $O(1)$ under refinement — it does not go to zero. What goes to zero is its *weighted* contribution, because it gets multiplied by the interpolation error. The estimator is small not because the residual is small, but because the residual is paired against something small.

In practice: the adaptive algorithm

① On some initial coarse mesh, compute the FEM solution.

② Calculate on each element the residual

$$E_j^2 = (x_j - x_{j-1})^2 \int_{x_{j-1}}^{x_j} \frac{1}{a(x)} R^2(U)\,\mathrm{d}x, \qquad \text{where} \qquad R(U) = f + (aU')'. \tag{1.140}$$

We want

$$E^2 = \sum_{j=1}^{M+1} E_j^2 \;\le\; \frac{\epsilon^2}{c^2}, \tag{1.141}$$

which is achieved if each element satisfies

$$E_j^2 \;\le\; \frac{\epsilon^2}{c^2(M+1)}. \tag{1.142}$$

③ Recalculate the solution on the new mesh, then repeat ② + ③.

Intuition: equidistribution

(1.142) splits the global budget ϵ^2/c^2 evenly over the $M+1$ elements. An element carrying more than its share gets refined; one carrying less is left alone. The converged mesh has roughly *equal error per element* — which is why adaptive meshes come out fine near a corner or a load and coarse in the quiet regions, without anyone specifying that by hand.
Note also the factor $(x_j - x_{j-1})^2$ in (1.140): halving an element cuts its indicator by four even at fixed residual, so refinement pays.

In mechanics terms

This is what a commercial solver's adaptive mesh refinement does. $R(U)$ is the equilibrium residual — how badly the computed stress field fails to balance the applied load inside each element — and E_j is the per-element error indicator you see coloured on the screen before remeshing.

► **Unclear in my notes:** The notes write $\|h^{-1}(e - \Pi_h e)\|_a \le C\|e'\|_a$ with $O(h^2)$ under the left side and $O(h)$ under the right. With the h^{-1} factor in front the left side should be $O(h)$, matching the right; the $O(h^2)$ presumably labels $e - \Pi_h e$ alone, before dividing by h. Check which grouping was meant.

1.10 CG(2); the beam equation and CG(3)

Lectures 10–11 (combined) Math 6660 — `missing.pdf` (colleague's notes)

These two lectures are missing from my own notebook; the transcription below is from a classmate's notes, so the wording is his, not mine. The material picks up mid-sentence from the adaptive criterion at the end of Lecture 9.

The big picture: Lectures 10–11

Two questions: what if $O(h)$ is not good enough? And what changes when the structure bends instead of stretches?

- **CG(2)**: add a midside node so each element carries a parabola. Rates improve to $O(h^2)$ in energy and $O(h^3)$ in L^2. The price is bandwidth — the matrix stops being tridiagonal.
- **The beam**, $u^{(iv)} = f$: four derivatives get shared two-and-two by integration by parts, so *both* u and v now need two derivatives. The space jumps from H^1 to H^2.
- That jump forbids kinks. Slope must match across element boundaries, so the elements must be C^1 and each node carries a deflection *and* a rotation.
- **CG(3)**: cubic Hermite shape functions — the classical beam element, derived from scratch.

Use it for: Why a bar element has one dof per node and a beam element has two, and why continuity requirements are a property of the *equation*, not a modelling choice.

Finishing the adaptive criterion

$$\text{If } E_j^2 \le \frac{\varepsilon^2}{c^2(M+1)} \text{ move to the next element; else refine.} \tag{1.143}$$

CG(2) method

Same problem as before,

$$\begin{cases} -(au')' = f & \text{on } (0,1), \\ u(0) = u(1) = 0, \end{cases} \tag{1.144}$$

with the same variational problem

$$\int_0^1 a\, u'v'\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x \qquad \forall\, v \in V, \tag{1.145}$$

$$V = \Big\{ v : \int_0^1 \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty,\ \ v(0) = v(1) = 0 \Big\}. \tag{1.146}$$

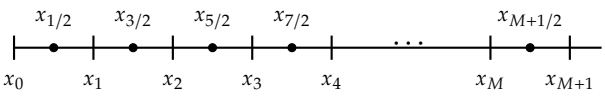
What changes is only the subspace. For CG(2),

$$V_h = \big\{ v \in V : \ v|_{[x_{j-1},x_j]} = a_j + b_j x + c_j x^2 \big\}. \tag{1.147}$$

Key: need to find a suitable basis for V_h.

Mesh $\mathcal{T}_h$: $0 = x_0 < x_1 < \cdots < x_M < x_{M+1} = 1$.

The simplest way to write CG(2) is to use Lagrangian polynomials. To do this, introduce a *submesh*:



The half-integer nodes need not be the midpoints, but we take them to be.

Intuition: what the extra node buys

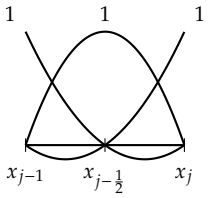
CG(1) fitted a straight line on each element, so two endpoint values pinned it down. A parabola has three coefficients, so it needs a third condition, and the cheapest place to put it is inside the element. The submesh node $x_{j-1/2}$ is not a new mesh point — the elements are unchanged — it is just a third *degree of freedom* carried by the same element.

In mechanics terms

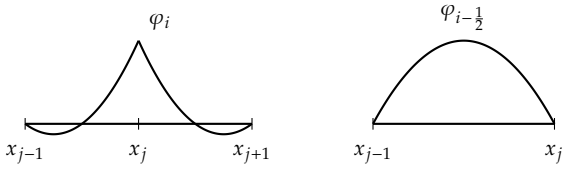
This is the three-node bar element. CG(1) is the 2-node linear bar with constant strain; CG(2) is the 3-node quadratic bar whose strain varies linearly along the element. The midside node is exactly the midside node of a Q8 or T6 element.

The basis

For a given element there are 3 Lagrange polynomials:



As *basis functions* over the whole mesh there are two shapes: the one sitting at an integer node, which spans two elements, and the bubble sitting at a half-integer node, which lives inside one element.



Explicitly,

$$\varphi_i(x) = \begin{cases} \dfrac{2\left(x - x_{i+\frac{1}{2}}\right)\left(x - x_{i+1}\right)}{\left(x_{i+1} - x_i\right)^2}, & x_i \le x \le x_{i+1}, \\ \dfrac{2\left(x - x_{i-\frac{1}{2}}\right)\left(x - x_{i-1}\right)}{\left(x_i - x_{i-1}\right)^2}, & x_{i-1} \le x \le x_i, \\ 0, & \text{otherwise,} \end{cases} \tag{1.148}$$

$$\varphi_{i-\frac{1}{2}}(x) = \begin{cases} \dfrac{4\left(x_i - x\right)\left(x - x_{i-1}\right)}{\left(x_i - x_{i-1}\right)^2}, & x_{i-1} \le x \le x_i, \\ 0, & \text{otherwise.} \end{cases} \tag{1.149}$$

Therefore

$$V_h = \operatorname{span}\left\{\varphi_{\frac{1}{2}}, \varphi_1, \varphi_{\frac{3}{2}}, \varphi_2, \ldots, \varphi_M, \varphi_{M+\frac{1}{2}}\right\}, \tag{1.150}$$

so if $U \in V_h$ then

$$U(x) = \sum_{j=1}^{2M+1} \xi_j \, \varphi_{j/2}(x). \tag{1.151}$$

Note. For any function $v \in V$,

$$\Pi_h v = \sum_{k=1}^{2M+1} v\left(x_{k/2}\right)\varphi_{k/2}(x), \tag{1.152}$$

and the test functions are $v = \varphi_{i/2}(x)$ for $i = 1, 2, \ldots, 2M+1$.

CG(2) variational problem

$$\sum_{j=1}^{2M+1} \xi_j \int_0^1 a(x)\, \varphi'_{i/2}\varphi'_{j/2}\, \mathrm{d}x = \int_0^1 f\, \varphi_{i/2}\, \mathrm{d}x. \tag{1.153}$$

This is a $(2M+1)\times(2M+1)$ symmetric positive definite system, *but not tridiagonal*.

Where this trips people up: the bandwidth changes, not just the size

Going from CG(1) to CG(2) is not simply "more unknowns." A basis function at an integer node now overlaps the bubbles on *both* neighbouring elements and the integer node two slots away, so the matrix picks up a second off-diagonal: it becomes pentadiagonal. Higher order buys accuracy per node and pays for it in bandwidth — the same trade a higher-order element makes in a commercial solver.

Example: $a \equiv 1$ and $h_j = h$ for all j

Stiffness matrix.

① $i = 1, 3, \ldots, 2M+1$ (the half-integer nodes):

$$A_{ii} = \int_{x_{\frac{i}{2}-\frac{1}{2}}}^{x_{\frac{i}{2}+\frac{1}{2}}} \left(\varphi'_{i/2}\right)^2 \mathrm{d}x = \frac{16}{3h}, \tag{1.154}$$

$$A_{(i-1,i)} = A_{(i,i-1)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \varphi'_{i/2}\, \varphi'_{\frac{i-1}{2}}\, \mathrm{d}x = -\frac{8}{3h}. \tag{1.155}$$

② $i = 2, 4, 6, \ldots, 2M$ (the integer nodes):

$$A_{ii} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \left[\varphi'_{i/2}\right]^2 \mathrm{d}x = \frac{14}{3h}, \tag{1.156}$$

$$A_{(i,i-1)} = A_{(i-1,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i}{2}}} \varphi'_{i/2}\, \varphi'_{\frac{i}{2}-\frac{1}{2}}\, \mathrm{d}x = -\frac{8}{3h}, \tag{1.157}$$

$$A_{(i,i-2)} = A_{(i-2,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{i/2}} \varphi'_{i/2}\, \varphi'_{\frac{i}{2}-1}\, \mathrm{d}x = \frac{1}{3h}. \tag{1.158}$$

RHS vector. Replace f by its interpolant,

$$f \approx \widetilde{\Pi}_h f = \sum_{j=0}^{2M+2} f\left(x_{j/2}\right)\varphi_{j/2}(x). \tag{1.159}$$

① $i = 1, 3, \ldots, 2M+1$:

$$f_i = \sum_{j=1}^{2M+1} f\left(x_{j/2}\right) \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \varphi_{j/2}\, \varphi_{i/2}\, \mathrm{d}x = \frac{h}{15}\left[f\left(x_{\frac{i-1}{2}}\right) + 8f\left(x_{i/2}\right) + f\left(x_{\frac{i+1}{2}}\right)\right]. \tag{1.160}$$

② $i = 2, 4, \dots, 2M$:

$$f_i = \frac{h}{30}\Big[-f\left(x_{\frac{i-2}{2}}\right) + 2f\left(x_{\frac{i-1}{2}}\right) + 8f\left(x_{i/2}\right) + 2f\left(x_{\frac{i+1}{2}}\right) - f\left(x_{\frac{i+2}{2}}\right)\Big]. \tag{1.161}$$

Therefore

$$\boldsymbol{A} = \frac{1}{3h}\begin{bmatrix} 16 & -8 & & & & & & \\ -8 & 14 & -8 & 1 & & & & \\ & -8 & 16 & -8 & & & & \\ & 1 & -8 & 14 & -8 & 1 & & \\ & & & \ddots & \ddots & \ddots & & \\ & & & 1 & -8 & 14 & -8 & \\ & & & & & -8 & 16 & \end{bmatrix}, \qquad \boldsymbol{Ax} = \boldsymbol{b}. \tag{1.162}$$

Note (energy norm error). Galerkin orthogonality gives $\|u - U\|_E \le \|u - v\|_E$ for all $v \in V_h$; taking $v = \Pi_h u$,

$$\boxed{\|u - U\|_E \le c\,\|h^2 u'''\|} \tag{1.163}$$

Note (L^2 error).

$$\boxed{\|u - U\| \le c\,\|h^3 u'''\|} \tag{1.164}$$

Intuition: what the exponents say

CG(1) gave $O(h)$ in energy and $O(h^2)$ in L^2; CG(2) gives $O(h^2)$ and $O(h^3)$. Each extra polynomial degree buys one extra power of h in both norms, and the L^2 rate stays one power ahead of the energy rate — the duality argument of Lecture 8 is what produces that extra power, and it does not care what degree the elements are.

Beam equation

$$\begin{cases} u^{(iv)} = f & \text{on } 0 < x < 1, \\ u(0) = u'(0) = u(1) = u'(1) = 0. \end{cases} \tag{1.165}$$

Variational problem

Multiply by v and integrate by parts twice:

$$\int_0^1 v\, u^{(iv)}\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x, \tag{1.166}$$

$$-\int_0^1 v' u'''\,\mathrm{d}x + \underbrace{\left[v\,u'''\right]_0^1}_{=0} = \int_0^1 f v\,\mathrm{d}x \qquad (\text{let } v(0) = v(1) = 0), \tag{1.167}$$

$$\int_0^1 v'' u''\,\mathrm{d}x - \underbrace{\left[v' u''\right]_0^1}_{=0} = \int_0^1 f v\,\mathrm{d}x \qquad (\text{let } v'(0) = v'(1) = 0), \tag{1.168}$$

so

$$\int_0^1 u'' v''\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x. \tag{1.169}$$

The space is

$$V = \left\{v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2\right]\mathrm{d}x < \infty,\ \ v(0) = v(1) = v'(0) = v'(1) = 0\right\}, \tag{1.170}$$

and functions in V are $C^1[0,1]$.

(Variational problem) Find $u \in V$ such that $\int_0^1 u''v''\,dx = \int_0^1 fv\,dx$ for all $v \in V$.

Intuition: why the space jumps from H^1 to H^2

The bar equation lost one derivative to the test function and kept one, so $\int (v')^2 < \infty$ was enough. The beam equation has four derivatives to share, two each, so now $\int (v'')^2 < \infty$ is required. That extra requirement is what forces C^1: a function with a corner has a jump in v', hence a delta in v'', hence $\int (v'')^2 = \infty$. Kinks were legal for the bar; they are illegal for the beam.

In mechanics terms

$\frac{1}{2}\int (u'')^2$ is the bending strain energy $\frac{1}{2}\int EI\,\kappa^2$ with $EI \equiv 1$, since u'' is the curvature. Requiring finite bending energy is requiring $u \in H^2$. And C^1 continuity is just the physical statement that a beam does not develop a kink under a distributed load — deflection *and* slope must match across an element boundary.

Energy minimization

Define

$$F(w) = \frac{1}{2}\int_0^1 (w'')^2\,dx - \int_0^1 fw\,dx. \tag{1.171}$$

(Minimization problem) Find $u \in V$ such that $F(u) \le F(w)$ for all $w \in V$.

Claim. (Var. Prob.) $\Longleftrightarrow$ (Min. Prob.)

($\Longrightarrow$) Assume $u \in V$ satisfies the variational problem. Let $w = u + v \in V$. Then

$$F(w) = F(u+v) = \underbrace{\frac{1}{2}\int_0^1 (u'')^2\,dx - \int_0^1 fu\,dx}_{=F(u)} + \frac{1}{2}\int_0^1 (v'')^2\,dx + \underbrace{\int_0^1 u''v''\,dx - \int_0^1 fv\,dx}_{=0 \text{ by } (1.169)}, \tag{1.172}$$

so

$$F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2\,dx \;\ge\; F(u). \tag{1.173}$$

Note. $F(w) = F(u) \;\Rightarrow\; v'' \equiv 0 \;\Rightarrow\; v \equiv 0$ (due to the BCs) $\Rightarrow\; u \equiv w$.

Intuition: why $v'' \equiv 0$ forces $v \equiv 0$ here

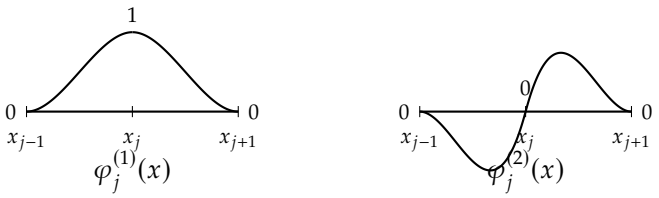
$v'' \equiv 0$ only says v is a straight line, $v = \alpha + \beta x$. It is the four boundary conditions in (1.170) that kill it: $v(0) = v'(0) = 0$ already force $\alpha = \beta = 0$. For the bar, the same step used $v' \equiv 0$ plus the two Dirichlet conditions. Same mechanism, one derivative higher.

CG(3) method

We need $V_h \subseteq V$. On each element $[x_{j-1}, x_j]$ we need a polynomial that matches *value and derivative* to its neighbours at x_{j-1} and x_j — that is 4 conditions, so we need a cubic (4 coefficients):

$$V_h = \left\{ v \in V : \; v|_{[x_{j-1},x_j]} = a_j + b_j x + c_j x^2 + d_j x^3 \right\}. \tag{1.174}$$

Key: find a basis for V_h. Two basis functions are attached to each node — one controlling the value, one controlling the slope:



They are fixed by the nodal table

	x_{j-1}	x_j	x_{j+1}
$\varphi_j^{(1)}$	0	1	0
$(\varphi_j^{(1)})'$	0	0	0
$\varphi_j^{(2)}$	0	0	0
$(\varphi_j^{(2)})'$	0	1	0

which gives

$$\varphi_j^{(1)} = \begin{cases} 3\left(\dfrac{x-x_{j-1}}{x_j-x_{j-1}}\right)^2 - 2\left(\dfrac{x-x_{j-1}}{x_j-x_{j-1}}\right)^3, & x_{j-1} \le x \le x_j, \\ 3\left(\dfrac{x-x_{j+1}}{x_j-x_{j+1}}\right)^2 - 2\left(\dfrac{x-x_{j+1}}{x_j-x_{j+1}}\right)^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}, \end{cases} \tag{1.175}$$

$$\varphi_j^{(2)} = \begin{cases} -\dfrac{1}{h_j}(x-x_{j-1})^2 + \dfrac{1}{h_j^2}(x-x_{j-1})^3, & x_{j-1} \le x \le x_j, \\ \dfrac{1}{h_{j+1}}(x-x_{j+1})^2 + \dfrac{1}{h_{j+1}^2}(x-x_{j+1})^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}. \end{cases} \tag{1.176}$$

CG(3) FEM: Find $U \in V_h$ such that $\displaystyle\int_0^1 U''v''\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x$ for all $v \in V_h$.

In mechanics terms

This is the Hermite beam element, exactly as it appears in a structural code. $\varphi_j^{(1)}$ and $\varphi_j^{(2)}$ are the cubic Hermite shape functions N_1, N_2, N_3, N_4; node j carries two degrees of freedom, a deflection and a rotation. The table above is the statement $N_i(\mathrm{dof}_k) = \delta_{ik}$ — each shape function switches on one dof and leaves the other three at zero. The C^1 requirement is why a beam element has a rotational dof at all, while a bar element does not.

► **Unclear in my notes:** In the nodal table for the Hermite basis, the entry for $(\varphi_j^{(1)})'$ at x_j is written in a way that reads as a 6 in the scan. It has to be 0 — (1.175) has a flat top at x_j — and I have transcribed it as 0. Worth a glance at the original.

► **Unclear in my notes:** Page 4 of the scan writes the two CG(2) error notes with u''' on both right-hand sides, $\|u-U\|_E \le c\|h^2u'''\|$ and $\|u-U\| \le c\|h^3u'''\|$. Check whether the second was meant to carry u''' or $u^{(4)}$.

1.11 CG(3) assembly: the Hermite stiffness matrix

Lecture 12 — Wednesday, September 23, 2026 Math 6660 — 12th.pdf

Lecture 12 re-states the beam problem from Lecture 11 and then carries the CG(3) discretisation all the way to the assembled linear system. Pages 7–9 of `missing.pdf` *cover the same material; the two sources agree throughout, and the transcription below follows my own notes.*

The big picture: Lecture 12

What does the beam stiffness matrix actually look like once assembled?

- Two kinds of unknown per node (a value, a slope) means every matrix entry is one of four pairings. The matrix splits into blocks: $\boldsymbol{A}$ deflection–deflection, $\boldsymbol{B}$ rotation–rotation, $\boldsymbol{C}$ the coupling.
- $\boldsymbol{C}$ is *anti*-symmetric — and the assembled $\boldsymbol{S}$ is still symmetric. Worth understanding rather than memorising.
- The load side now needs **two** vectors: a distributed load lumps into nodal forces *and* nodal moments.
- The numbers that come out — $12/h^3, 6/h^2, 4/h, 2/h$ — are the textbook Euler–Bernoulli element matrix with $EI = 1$.

Use it for: Recognising the beam element you already know, written in the mathematics department's notation.

Recap: the beam problem and its CG(3) space

$$\begin{cases} u^{(iv)} = f & \text{in } (0,1), \\ u(0) = u(1) = u'(0) = u'(1) = 0, \end{cases} \tag{1.177}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x < \infty, \;\; v(0) = v(1) = v'(0) = v'(1) = 0 \right\}. \tag{1.178}$$

Functions in V are C^1. Find $u \in V$ such that

$$\int_0^1 u'' v'' \, \mathrm{d}x = \int_0^1 f v \, \mathrm{d}x \qquad \forall\, v \in V. \tag{1.179}$$

FEM: $V_h \subset V$, spanned by the two Hermite shapes per node — $\varphi_j^{(1)}$, which carries the nodal value, and $\varphi_j^{(2)}$, which carries the nodal slope (Lecture 11, (1.175)–(1.176)):

$$V_h = \operatorname{span}\{\varphi_1^{(1)}, \dots, \varphi_M^{(1)}, \; \varphi_1^{(2)}, \dots, \varphi_M^{(2)}\}, \qquad \dim V_h = 2M. \tag{1.180}$$

Trial functions:

$$U = \sum_{j=1}^{M} \xi_j \, \varphi_j^{(1)}(x) + \sum_{j=1}^{M} \eta_j \, \varphi_j^{(2)}(x). \tag{1.181}$$

Test functions: $v = \varphi_1^{(1)}, \dots, \varphi_M^{(1)}, \varphi_1^{(2)}, \dots, \varphi_M^{(2)}$.

The interpolant and its errors

If $u \in V$, then

$$\Pi_h u = \sum_{j=1}^{M} u(x_j)\,\varphi_j^{(1)}(x) + \sum_{j=1}^{M} u'(x_j)\,\varphi_j^{(2)}(x), \qquad \text{where } \Pi_h u \in V_h. \tag{1.182}$$

Interpolation errors:

$$\|u - \Pi_h u\|_{L^2(0,1)} \le c_1 \|h^4 u^{(4)}\|_{L^2(0,1)}, \tag{1.183}$$

$$\|(u - \Pi_h u)'\|_{L^2(0,1)} \le c_2 \|h^3 u^{(4)}\|_{L^2(0,1)}, \tag{1.184}$$

$$\|(u - \Pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}. \tag{1.185}$$

Intuition: why $u^{(4)}$, and why the exponents step down

The Hermite interpolant matches value and slope at both ends of every element, so it reproduces *every cubic exactly*. The first thing it cannot reproduce is the fourth derivative — which is why $u^{(4)}$, and nothing lower, sits on every right-hand side. Each derivative you take of the error costs one power of h: h^4 for the function, h^3 for the slope, h^2 for the curvature. Since the energy norm for the beam measures v'', (1.185) is the one that controls the FEM error, giving $O(h^2)$.

Example: uniform mesh, $h_j = h$ for all j

With

$$\int_0^1 U''v''\,\mathrm{d}x = \int_0^1 f v\,\mathrm{d}x, \qquad v = \varphi_1^{(1)}, \dots, \varphi_M^{(1)}, \varphi_1^{(2)}, \dots, \varphi_M^{(2)}, \tag{1.186}$$

the stiffness matrix comes out of four families of integrals. Testing against $\varphi_k^{(1)}$, $k = 1, 2, \dots, M$:

$$\int_0^1 U''v''\,\mathrm{d}x = \sum_{j=1}^{M} \xi_j \int_0^1 (\varphi_j^{(1)})''(\varphi_k^{(1)})''\,\mathrm{d}x + \sum_{j=1}^{M} \eta_j \int_0^1 (\varphi_j^{(2)})''(\varphi_k^{(1)})''\,\mathrm{d}x, \tag{1.187}$$

and then again, testing against $\varphi_k^{(2)}$, $k = 1, 2, \dots, M$:

$$\sum_{j=1}^{M} \xi_j \int_0^1 (\varphi_j^{(1)})''(\varphi_k^{(2)})''\,\mathrm{d}x + \sum_{j=1}^{M} \eta_j \int_0^1 (\varphi_j^{(2)})''(\varphi_k^{(2)})''\,\mathrm{d}x. \tag{1.188}$$

That is a block matrix

$$\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline -\boldsymbol{C} & \boldsymbol{B} \end{array}\right] \in \mathbb{R}^{2M\times 2M}, \qquad \text{here } \boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{M\times M}. \tag{1.189}$$

Intuition: where the four blocks come from

There are two kinds of unknown (ξ, a value; η, a slope) and two kinds of test function, so every entry of the matrix is one of four pairings: value–value, value–slope, slope–value, slope–slope. $\boldsymbol{A}$ is the first, $\boldsymbol{B}$ the last, $\boldsymbol{C}$ the cross terms. Nothing about the physics changed — it is still $\int u''v''$ — only the bookkeeping, because each node now carries two numbers.

The three blocks

Now,

$$A_{jk} = \int_0^1 (\varphi_j^{(1)})''(\varphi_k^{(1)})'' \, \mathrm{d}x, \qquad \boldsymbol{A} = \frac{12}{h^3}\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix}, \tag{1.190}$$

which is $M \times M$, tridiagonal, symmetric.

$$B_{jk} = \int_0^1 (\varphi_j^{(2)})''(\varphi_k^{(2)})'' \, \mathrm{d}x, \qquad \boldsymbol{B} = \frac{2}{h}\begin{bmatrix} 4 & 1 & & \\ 1 & 4 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & 4 \end{bmatrix}, \tag{1.191}$$

which is $M \times M$, tridiagonal, symmetric.

$$C_{jk} = \int_0^1 (\varphi_j^{(1)})''(\varphi_k^{(2)})'' \, \mathrm{d}x, \qquad \boldsymbol{C} = \frac{6}{h^2}\begin{bmatrix} 0 & 1 & & \\ -1 & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & -1 & 0 \end{bmatrix}, \tag{1.192}$$

which is $M \times M$, tridiagonal — but *not* symmetric. It is anti-symmetric:

$$\boldsymbol{C}^T = -\boldsymbol{C}. \tag{1.193}$$

Therefore

$$\boldsymbol{S} = \left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline \boldsymbol{C}^T & \boldsymbol{B} \end{array}\right], \qquad \boldsymbol{S}^T = \boldsymbol{S}, \tag{1.194}$$

so $\boldsymbol{S}$ is $2M \times 2M$ and symmetric.

Where this trips people up: an anti-symmetric block does not break symmetry

$\boldsymbol{C}^T = -\boldsymbol{C}$ is exactly what makes $\boldsymbol{S}$ symmetric: the lower-left block *is* $\boldsymbol{C}^T$, so writing it as $-\boldsymbol{C}$ in (1.189) is the same statement. The minus sign is a consequence of the sign convention for the slope degree of freedom, not a loss of symmetry — $\int u''v''$ is self-adjoint, so the matrix could not have come out unsymmetric.

In mechanics terms

The $12/h^3$, $6/h^2$, $4/h$, $2/h$ pattern in (1.190)–(1.192) is the classical Euler–Bernoulli beam element stiffness matrix with $EI = 1$. In a structural code the element matrix is written once as $\frac{EI}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$ and assembled; here the same numbers arrive sorted by *type* of degree of freedom instead of by node, which is why they appear as three separate blocks.

RHS

Interpolate the load with the same Hermite basis,

$$f \longrightarrow \widetilde{\Pi}_h f = \sum_{j=0}^{M+1} f(x_j)\,\varphi_j^{(1)}(x) + \sum_{j=0}^{M+1} f'(x_j)\,\varphi_j^{(2)}(x), \tag{1.195}$$

$$\int_0^1 f v \,\mathrm{d}x \;\longrightarrow\; \int_0^1 (\widetilde{\Pi}_h f) v \,\mathrm{d}x. \tag{1.196}$$

Now, for $j = 1, 2, \ldots, M$,

$$f_j^{(1)} = \frac{h}{70}\Big[9f(x_{j-1}) + 52f(x_j) + 9f(x_{j+1})\Big] + \frac{13h^2}{420}\Big[f'(x_{j-1}) - f'(x_{j+1})\Big], \tag{1.197}$$

$$f_j^{(2)} = \frac{13h^2}{420}\Big[f(x_{j+1}) - f(x_{j-1})\Big] - \frac{h^3}{420}\Big[3f'(x_{j-1}) - 8f'(x_j) + 3f'(x_{j+1})\Big]. \tag{1.198}$$

The final system

$$\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline -\boldsymbol{C} & \boldsymbol{B} \end{array}\right] \left[\begin{array}{c} \boldsymbol{\xi} \\ \hline \boldsymbol{\eta} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{f}^{(1)} \\ \hline \boldsymbol{f}^{(2)} \end{array}\right], \tag{1.199}$$

which is $2M \times 2M$, symmetric, positive definite. Here $\boldsymbol{\eta}$ is the vector of *derivative values at the nodes*.

In mechanics terms

(1.199) is the assembled beam problem in the form a structural solver produces it: $\boldsymbol{\xi}$ are the nodal deflections, $\boldsymbol{\eta}$ the nodal rotations, $\boldsymbol{f}^{(1)}$ the consistent nodal forces and $\boldsymbol{f}^{(2)}$ the consistent nodal moments. The $13h^2/420$ and $h^3/420$ coefficients in (1.197)–(1.198) are the consistent load vector for a distributed load on a Hermite beam element — the same numbers a code computes when you apply a pressure rather than a point load.

Translation: beam FEM

Math department writes	You should hear
ξ_j, the coefficient of $\varphi_j^{(1)}$	the nodal deflection w_j
η_j, the coefficient of $\varphi_j^{(2)}$	the nodal rotation θ_j
$\boldsymbol{A}$ block	deflection–deflection stiffness
$\boldsymbol{B}$ block	rotation–rotation stiffness
$\boldsymbol{C}$ block	the deflection–rotation coupling
$\boldsymbol{S}$ symmetric positive definite	the structure is stable and the energy is a genuine minimum
$\widetilde{\Pi}_h f$	lumping the distributed load onto nodal forces and moments

► **Unclear in my notes:** On page 2 the first interpolation error is written $\|u - \Pi_h u\| \le c_1 \|h^4 u^4\|$ — the exponent on u should be the superscript (4), matching the other two lines; transcribed as $u^{(4)}$. Similarly the second sum in $\Pi_h u$ on that page runs to n rather than M.

1.12 The abstract framework: Lax–Milgram, Céa, and the move to 2D

Lectures 13–18 — late September to mid-October, 2026 Math 6660 — `prashant2.pdf` (colleague's notes)

*These lectures are missing from my own notebook; the transcription below follows a classmate's notes, so the wording and the order of presentation are his, not mine. Only two dates are legible in the scan — **2/Oct/26** at the abstract variational formulation, and **12/Oct/26** at the biharmonic example — so the lecture numbering*

above is my reconstruction, not something written in the source. Pages 1–18 of that scan repeat material already transcribed in Lectures 7–12; the transcription starts where the new material does.

The big picture: Lectures 13–18

Every problem so far has been solved by the same four moves. What is the general theorem, and what does it cost to go to two dimensions?

- The bar and the beam get rewritten in *one* language: $B(u,v) = L(v)$. Everything specific to the problem is pushed into three properties of B — symmetric, bounded, elliptic.
- **Lax–Milgram** then says: those three properties alone guarantee a unique solution. No more re-deriving existence for each new PDE.
- **Céa's lemma** in general form: $\|u - u_h\| \le (\beta^*/\beta_*)\|u - v\|$. The constant 1 we had before was the lucky case.
- The same abstract theorem gives, for free, that the stiffness matrix is **symmetric positive definite** — in any dimension, any polynomial order, any mesh.
- Then 2D: Poisson on a triangulated domain, the biharmonic plate (and why it needs the 21-degree-of-freedom Argyris element), and how a mesh is actually generated — Voronoi, Delaunay, level sets.

Use it for: This is the point where the course stops being about one-dimensional examples and becomes the theory a real FE code is built on.

Part I — the abstract formulation

So far: two examples

Both problems solved so far have the same shape.

(1) The two-point BVP.

$$-(au')' = f \ \text{ in } (0,1), \qquad u(0) = u(1) = 0, \tag{1.200}$$

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 \right] \mathrm{d}x < \infty, \ \ v(0) = v(1) = 0 \right\}, \tag{1.201}$$

$$B(u,v) = \int_0^1 a\, u'v' \,\mathrm{d}x, \qquad L(v) = \int_0^1 f v \,\mathrm{d}x. \tag{1.202}$$

(2) The beam equation.

$$V = \left\{ v : \int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x < \infty, \ \ v(0) = v(1) = v'(0) = v'(1) = 0 \right\}, \tag{1.203}$$

$$B(u,v) = \int_0^1 u''v'' \,\mathrm{d}x, \qquad L(v) = \int_0^1 f v \,\mathrm{d}x. \tag{1.204}$$

In both cases:

$$\textbf{(Var. Prob.)} \quad \text{Find } u \in V \text{ s.t. } B(u,v) = L(v) \quad \forall\, v \in V, \tag{1.205}$$

$$F(w) = \tfrac{1}{2}B(w,w) - L(w), \qquad \textbf{(Energy Min.)} \quad \text{Find } u \in V \text{ s.t. } F(u) \le F(w) \ \ \forall\, w \in V. \tag{1.206}$$

Note. We seek a **conforming** FEM: $V_h \subset V$. (A non-conforming FEM has $V_h \not\subset V$.)

Intuition: what is actually being abstracted

Nothing new is being claimed here — (1.202) and (1.204) are the weak forms already derived. The move is to stop writing the integrals and give them a name. Once the bar and the beam are both "$B(u,v) = L(v)$", any theorem proved about B and L applies to both at once, and to the 2D problems later in these lectures that have not been written down yet. The price is that B and L say nothing about *which* problem you are solving. Everything that distinguishes a bar from a beam has been pushed into the space V and into the three properties of B defined below.

Multi-index notation

Consider a domain $\Omega \subset \mathbb{R}^d$. Derivatives are defined via a **multi-index**: let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$ be a d-tuple, where the α_i are non-negative integers. The *length* of α is

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d, \tag{1.207}$$

and the corresponding derivative of v is

$$D^\alpha v = \frac{\partial^{|\alpha|} v}{\partial x_1^{\alpha_1}\, \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}. \tag{1.208}$$

Ex. If $d = 2$ and $|\alpha| = 2$:

$$D^{(2,0)}v = \frac{\partial^2 v}{\partial x_1^2}, \qquad D^{(0,2)}v = \frac{\partial^2 v}{\partial x_2^2}, \qquad D^{(1,1)}v = \frac{\partial^2 v}{\partial x_1 \partial x_2}.$$

Intuition: why bother

In 1D you can write v, v', v'' and be done. In 2D "the second derivatives" is already three different objects, and in d dimensions it is $\binom{d+1}{2}$ of them. The multi-index is bookkeeping that lets "*all* derivatives up to order r" be written as a single sum $\sum_{|\alpha|\le r}$ instead of an expanding list. Every Sobolev definition below uses exactly that.

Sobolev spaces

$$L^1_{\text{loc}}(\Omega) := \big\{ v : \ v \in L^1(K) \text{ for any compact subset } K \text{ inside } \Omega \big\}. \tag{1.209}$$

Defn (Sobolev space).

$$W^{r,q}(\Omega) = \Big\{ v \in L^1_{\text{loc}}(\Omega) : \ \|v\|_{W^{r,q}(\Omega)} < \infty \Big\}, \tag{1.210}$$

where

$$\|v\|_{W^{r,q}(\Omega)} := \Big[\sum_{|\alpha| \le r} \|D^\alpha v\|^q_{L^q(\Omega)} \Big]^{1/q}, \qquad 1 \le q < \infty, \quad r \ge 0, \tag{1.211}$$

and

$$\|v\|^q_{L^q(\Omega)} = \int_\Omega |v|^q \, \mathrm{d}x. \tag{1.212}$$

Defn (Sobolev space with homogeneous BCs).

$$W^{r,q}_0(\Omega) = \Big\{ v \in W^{r,q}(\Omega) : \ D^\alpha v\big|_{\partial\Omega} = 0 \text{ for } |\alpha| < r \Big\}. \tag{1.213}$$

► **Unclear in my notes:** On the page where the L^q norm is first introduced my colleague's notes write $\|u\|_{L^q(\Omega)} := \int_\Omega |u| \, \mathrm{d}x$, with no exponent q and no outer root. That cannot be right — it would make the L^q norm independent of q — and the correct form appears two lines later as (1.212). Transcribed as (1.212); worth confirming against the lecturer's own notes.

Hilbert spaces

Defn (Hilbert space). When $q = 2$ we write $W^{r,2}(\Omega) = H^r(\Omega)$. This is called a **Hilbert space**, with

$$\|v\|_{H^r(\Omega)} = \sqrt{(v,v)_{H^r(\Omega)}}, \qquad (v,w)_{H^r(\Omega)} \coloneqq \sum_{|\alpha|\le r} \int_\Omega \big(D^\alpha v\big)\big(D^\alpha w\big)\,\mathrm{d}x. \tag{1.214}$$

Note. $r = 0$ also makes sense: $H^0(\Omega) = L^2(\Omega)$.

Defn (semi-norm).

$$|v|_{H^r(\Omega)} = \sqrt{\sum_{|\alpha|=r} \int_\Omega \big(D^\alpha v\big)^2\,\mathrm{d}x}. \tag{1.215}$$

Intuition: norm vs. semi-norm, in one line

The norm sums over $|\alpha| \le r$ — *every* derivative up to order r, including the function itself. The semi-norm sums over $|\alpha| = r$ — only the *top* ones. So $|v|_{H^r} = 0$ does not force $v = 0$; it only forces v to be a polynomial of degree $< r$. That is why a semi-norm appears on the right-hand side of every interpolation estimate: what the interpolant cannot reproduce is exactly the top derivative, and the lower ones are irrelevant to the error.

Translation: the spaces in engineering terms

Math department writes	You should hear
$L^2(\Omega)$	finite "size": $\int v^2 < \infty$
$H^1(\Omega)$	finite strain energy for a bar: $\int (v')^2 < \infty$
$H^1_0(\Omega)$	the same, and clamped at the boundary
$H^2_0(\Omega)$	finite bending energy, clamped *and* zero slope — the beam/plate space
$\|v\|_{H^r}$	the energy-like part only, with the rigid-body content stripped out

Note. For the remainder of the discussion, let V be a Hilbert space with inner product $(\cdot,\cdot)_V$ and corresponding norm $\|\cdot\|_V$.

Linear functionals and the dual space

Defn. A mapping $L : V \to \mathbb{R}$ is a **linear functional** if

$$L(\alpha u + \beta v) = \alpha L(u) + \beta L(v) \qquad \forall\, \alpha,\beta \in \mathbb{R},\ \forall\, u,v \in V. \tag{1.216}$$

L is **bounded** in the norm $\|\cdot\|_V$ if there exists a constant $\widetilde{L} > 0$ such that

$$|L(v)| \le \widetilde{L}\,\|v\|_V \qquad \forall\, v \in V. \tag{1.217}$$

Note. A bounded linear functional is continuous:

$$L(v) - L(w) = L(v - w) \implies |L(v) - L(w)| = |L(v - w)| \le \widetilde{L}\|v - w\|_V.$$

The converse is also true: a continuous linear functional is bounded.

Defn. The set of bounded linear functionals on V is termed the **dual space** of V, denoted V', equipped with the norm

$$\|L\|_{V'} = \sup_{\substack{v\in V\\ v\neq 0}} \frac{L(v)}{\|v\|_V} \;\leq\; \widetilde{L}. \tag{1.218}$$

Ex. (dual space of $L^q(\Omega)$). Consider the linear functional $L(v) = \int_\Omega f(x)v(x)\,\mathrm{d}x$ for $f \in L^{q'}(\Omega)$ where $\frac{1}{q} + \frac{1}{q'} = 1$. **Hölder's inequality**,

$$\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^{q'}(\Omega)}\|v\|_{L^q(\Omega)} \qquad \forall\, u \in L^{q'}(\Omega),\ v \in L^q(\Omega), \tag{1.219}$$

gives $|L(v)| \leq \|f\|_{L^{q'}}\|v\|_{L^q}$. So every $f \in L^{q'}(\Omega)$ can be viewed as a bounded linear functional on $L^q(\Omega)$; every bounded linear functional on $L^q(\Omega)$ is of this form; hence $L^{q'}(\Omega)$ is the dual space of $L^q(\Omega)$.

Thm (Riesz representation). Let H be a Hilbert space with scalar product $(\cdot,\cdot)_H$. Then for any continuous linear functional $L : H \to \mathbb{R}$ there exists a *unique* $u \in H$ such that

$$L(v) = (u, v)_H \qquad \forall\, v \in H. \tag{1.220}$$

In mechanics terms: what Riesz is doing for us

A load is a functional: it eats a virtual displacement and returns a number (the work done). Riesz says that any such load, provided it is bounded, is secretly *a displacement field* paired against the virtual displacement through the energy inner product. That identification is the entire existence proof for Lax–Milgram below — the load L gets converted into the solution u.

Abstract variational problems

Defn (bilinear form). Let V be a Hilbert space with scalar product $(\cdot,\cdot)_V$. Then $B(\cdot,\cdot) : V\times V \to \mathbb{R}$ is a **bilinear form** if

$$B(u, \alpha v + \beta w) = \alpha B(u, v) + \beta B(u, w), \qquad B(\alpha u + \beta v, w) = \alpha B(u, w) + \beta B(v, w) \tag{1.221}$$

for all $\alpha, \beta \in \mathbb{R}$ and all $u, v, w \in V$.

The three properties that everything below depends on:

(S) Symmetric. $B(u, v) = B(v, u) \quad \forall\, u, v \in V$.

(CB) Continuous / bounded. There exists $\beta^* > 0$ such that

$$B(u, v) \leq |B(u, v)| \leq \beta^*\|u\|_V\|v\|_V \qquad \forall\, u, v \in V.$$

(VE) V-elliptic (a.k.a. coercive). There exists $\beta_* > 0$ such that

$$B(v, v) \geq \beta_*\|v\|_V^2 \qquad \forall\, v \in V.$$

Defn (abstract variational problem). Let $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a bilinear form satisfying (S), (CB) and (VE). Let $L(\cdot) : V \to \mathbb{R}$ be a linear functional bounded in $\|\cdot\|_V$, i.e. $|L(v)| \leq \widetilde{L}\|v\|_V\ \forall v \in V$. Then

$$\textbf{(1) Var. Prob.} \quad \text{Find } u \in V \text{ s.t. } B(u, v) = L(v)\ \ \forall\, v \in V, \tag{1.222}$$

$$\textbf{(2) Energy Min.} \quad \text{Find } u \in V \text{ s.t. } F(u) \leq F(w)\ \ \forall\, w \in V, \quad F(v) = \tfrac{1}{2}B(v, v) - L(v). \tag{1.223}$$

In mechanics terms: the three conditions as structural statements

(S) is Maxwell–Betti reciprocity: the work done by load A through displacement B equals the work of B through A. It is what makes $\boldsymbol{K}$ symmetric.
(CB) says the structure cannot store infinite energy from a finite displacement — the material is bounded in stiffness.
(VE) says every admissible non-zero displacement stores *positive* energy, bounded below by β_*. This is the statement that the structure has no mechanism and no rigid-body mode left: it is why $\boldsymbol{K}$ is positive definite and not merely positive semi-definite. β_* is, in effect, the softest mode of the structure.

Claim: (1) and (2) are equivalent

($\Longrightarrow$) Assume $u \in V$ satisfies the variational problem. Let $w = u + v \in V$ for some $v \in V$. Then

$$F(w) = F(u+v) = \tfrac{1}{2}B(u+v, u+v) - L(u+v),$$
$$B(u+v, u+v) = B(u, u+v) + B(v, u+v) = B(u,u) + 2B(u,v) + B(v,v) \quad \text{(used (S))}, \tag{1.224}$$

so

$$F(u+v) = \underbrace{\left[\tfrac{1}{2}B(u,u) - L(u)\right]}_{F(u)} + \underbrace{\left[B(u,v) - L(v)\right]}_{=0 \text{ by (Var. Prob.)}} + \tfrac{1}{2}B(v,v), \tag{1.225}$$

i.e. $F(w) = F(u) + \frac{1}{2}B(v,v)$. But $B(v,v) \geq \beta_* \|v\|_V^2 > 0$ for $v \neq 0$ by (VE), hence

$$F(u) \leq F(w) \qquad \forall\, w \in V. \tag{1.226}$$

($\Longleftarrow$) Assume $u \in V$ satisfies $F(u) \leq F(w)\ \forall w \in V$. Let

$$g(\varepsilon) = F(u + \varepsilon v) \qquad \text{for fixed but arbitrary } v \in V. \tag{1.227}$$

Expanding,

$$g(\varepsilon) = F(u) + \varepsilon B(u,v) - \varepsilon L(v) + \tfrac{\varepsilon^2}{2}B(v,v), \tag{1.228}$$
$$g'(\varepsilon) = B(u,v) - L(v) + \varepsilon\, B(v,v), \tag{1.229}$$

and $g'(0) = 0$ gives $B(u,v) = L(v)$. □

Recall: this is Lecture 3 and Lecture 11 again

The bar version of this proof is Lecture 3; the beam version is Lecture 11. The algebra is identical in all three — complete the square one way, take $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}$ at $\varepsilon = 0$ the other. What is new is only that it is now done *once*, for every problem of this shape at the same time.

The Lax–Milgram theorem [2/Oct/26]

Thm (Lax–Milgram). Let V be a Hilbert space with norm $\|\cdot\|_V$. Let $B(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a bilinear form satisfying (S), (CB) and (VE). Let $L(\cdot) : V \to \mathbb{R}$ be a bounded linear functional. Then there exists a **unique** solution to the variational problem

$$\text{find } u \in V \text{ s.t. } B(u,v) = L(v)\ \ \forall\, v \in V,$$

with the property that

$$\|u\|_V \leq \frac{\widetilde{L}}{\beta_*},$$

where $\widetilde{L}, \beta_*$ are real positive constants.

Proof. Since $B(\cdot,\cdot)$ is symmetric and V-elliptic, it induces a scalar product on V:

$$[u,v] := B(u,v) \qquad \forall\, u,v \in V. \tag{1.230}$$

By (VE) and (CB),

$$[v,v] = B(v,v) \geq \beta_* \|v\|_V^2, \qquad [v,v] = B(v,v) \leq \beta^* \|v\|_V^2, \tag{1.231}$$

so

$$\boxed{\beta_* \|v\|_V^2 \leq [v,v] \leq \beta^* \|v\|_V^2} \tag{1.232}$$

and the norms $\sqrt{[v,v]}$ and $\|v\|_V$ are **equivalent**. Therefore $L(\cdot)$ is bounded in both. By the Riesz representation theorem (1.220) there exists a unique $u \in V$ such that $L(v) = [u,v]\ \forall v \in V$, i.e. a unique $u \in V$ with $B(u,v) = L(v)\ \forall v \in V$.

For the solution bound,

$$\beta_* \|u\|_V^2 \leq B(u,u) = L(u) \leq \widetilde{L}\|u\|_V \implies \|u\|_V \leq \frac{\widetilde{L}}{\beta_*}. \quad \square \tag{1.233}$$

Intuition: what Lax–Milgram buys, in one sentence

Up to now, every new BVP needed its own argument that a solution exists and is unique. Lax–Milgram replaces all of them with a checklist: write the weak form, verify (S), (CB), (VE), and you are done — existence, uniqueness, and a bound on how big the answer can get, all three.
The bound (1.233) is also the *stability* statement: double the load and the response at most doubles, with constant $1/\beta_*$. A structure with a nearly-singular mode has a tiny β_* and is predicted, correctly, to be sensitive.

Where this trips people up: (S) is not actually needed for Lax–Milgram

The theorem as stated in lecture assumes (S), (CB) *and* (VE), and the proof given uses (S) to build the inner product $[u,v]$ and then quotes Riesz. The standard Lax–Milgram theorem does *not* require symmetry — it holds for non-symmetric B, at the cost of a longer proof that cannot use Riesz directly. Symmetry is what lets the short proof above work, and it is also what gives the energy-minimisation reading (1.223); a non-symmetric problem (e.g. one with a $b\,u'$ convection term, cf. Lecture 1) has no energy functional to minimise but still has a unique weak solution.

Part II — conforming FEM in the abstract setting

The discrete problem

Replace V by a finite-dimensional subspace $V_h \subset V$ (*conforming*):

$$\text{Find } u_h \in V_h \text{ s.t. } B(u_h,v) = L(v) \quad \forall\, v \in V_h. \tag{1.234}$$

Lax–Milgram still applies to this, so there exists a unique solution with $\|u_h\|_V \leq \widetilde{L}/\beta_*$.

Let $\{\varphi_i\}_{i=1}^M$ be a basis for V_h, take $v = \varphi_j$, $j = 1,\dots,M$, and write $u_h = \sum_{i=1}^M \xi_i \varphi_i$. Then

$$\sum_{i=1}^{M} B(\varphi_i,\varphi_j)\,\xi_i = L(\varphi_j), \qquad j = 1,2,\dots,M, \tag{1.235}$$

i.e. the linear system

$$\boxed{\boldsymbol{A}\,\boldsymbol{\xi} = \boldsymbol{b}} \qquad A_{ij} = B(\varphi_i,\varphi_j), \quad \boldsymbol{\xi} = (\xi_1,\xi_2,\dots,\xi_M), \quad b_j = L(\varphi_j). \tag{1.236}$$

Claim: under (S) and (VE), $\boldsymbol{A}$ is symmetric positive definite

(1) Symmetry follows directly from (S):

$$A_{ij} = B(\varphi_i, \varphi_j) = B(\varphi_j, \varphi_i) = A_{ji}. \tag{1.237}$$

(2) Positive definiteness. Let $w = \sum_{i=1}^{M} w_i \varphi_i \in V_h$. By (VE), $B(w,w) \geq \beta_* \|w\|_V^2$, so if $w \not\equiv 0$ then $B(w,w) > 0$. Expanding,

$$B\Big(\sum_{i=1}^{M} w_i\varphi_i,\, w\Big) = \sum_{i=1}^{M} w_i B(\varphi_i, w) = \sum_{i=1}^{M} w_i B\Big(\varphi_i, \sum_{j=1}^{M} w_j \varphi_j\Big) = \sum_{i=1}^{M}\sum_{j=1}^{M} w_i \underbrace{B(\varphi_i,\varphi_j)}_{A_{ij}} w_j = \boldsymbol{w}^T \boldsymbol{A}\, \boldsymbol{w} > 0 \tag{1.238}$$

by (VE). Hence $\boldsymbol{A}$ is positive definite. □

Note. This means a conforming FEM for this abstract variational problem will generate an SPD linear system — *independent of* the number of dimensions, the polynomial order, the shape of the solution domain Ω, the type of mesh, etc.

Intuition: why this note is the point of the whole abstraction

Every stiffness matrix met so far came out symmetric positive definite, and each time it was verified by looking at the numbers. (1.238) shows it was never a coincidence: $\boldsymbol{w}^T \boldsymbol{A}\boldsymbol{w}$ *is* $B(w,w)$, the energy of the function w, so (VE) — a statement about the physics — becomes positive definiteness, a statement about the matrix. Nothing about 1D, cubics or uniform meshes entered the argument, which is exactly why the result survives into 2D in Part III.

Céa's lemma

Thm (Céa's lemma). Let V be a Hilbert space with norm $\|\cdot\|_V$, let B be a bilinear form satisfying (S), (CB) and (VE), and let L be a linear bounded functional. Let $V_h \subset V$ be a finite-dimensional subspace, and let u, u_h solve (1.222) and (1.234) respectively. Then

$$\|u - u_h\|_V \;\leq\; \frac{\beta^*}{\beta_*}\|u - v\|_V \qquad \forall\, v \in V_h.$$

Proof. Subtract (Var. Prob.) and (FEM):

$$B(u,v) - B(u_h,v) = 0 \;\; \forall\, v \in V_h \qquad \Longrightarrow \qquad \boxed{B(u - u_h,\, v) = 0 \quad \forall\, v \in V_h} \tag{1.239}$$

which is **Galerkin orthogonality**. Now take

$$B(u-u_h,\, u-u_h) = B(u-u_h,\, u - v + v - u_h) = B(u-u_h,\, u-v) + \underbrace{B(u-u_h,\, v-u_h)}_{=0 \text{ by (1.239)}}, \tag{1.240}$$

so, using (VE) on the left and (CB) on the right,

$$\beta_*\|u-u_h\|_V^2 \leq B(u-u_h, u-u_h) = B(u-u_h, u-v) \leq \beta^* \|u-u_h\|_V \|u-v\|_V, \tag{1.241}$$

and cancelling one factor $\|u-u_h\|_V$,

$$\boxed{\|u-u_h\|_V \leq \frac{\beta^*}{\beta_*}\|u-v\|_V} \qquad \square \tag{1.242}$$

Note (error estimate in $\|\cdot\|_V$). Take $v = \mathcal{I}_h u$, the projection of $u \in V$ into V_h. Then

$$\|u - u_h\|_V \le \frac{\beta^*}{\beta_*} \|u - \mathcal{I}_h u\|_V, \tag{1.243}$$

where $\|u - \mathcal{I}_h u\|_V$ is the *interpolation* error measured in $\|\cdot\|_V$. If you can generate a sequence of meshes and associated bases such that $\|u - \mathcal{I}_h u\|_V \to 0$ as $h \to 0$, then you have convergence; typically

$$\|u - \mathcal{I}_h u\|_V = O(h^p). \tag{1.244}$$

Where this trips people up: the constant is no longer 1

Lecture 6 proved $\|u - U\|_E \le \|u - v\|_E$ — constant exactly 1 — and that is *not* the general case. It happened because the energy norm there was built from the same a as the bilinear form, making $\beta^* = \beta_* = 1$ by construction. In a general norm $\|\cdot\|_V$ the constant is β^*/β_*, a condition-number sort of quantity: u_h is *quasi*-optimal, within a fixed factor of the best approximation, not the best approximation itself.

Example: the 1D Dirichlet problem, checked against the three conditions

$$-u'' = f \ \text{ in } (0,1), \qquad u(0) = u(1) = 0, \qquad I = (0,1), \qquad V = H_0^1(I), \tag{1.245}$$

$$\|v\|_V^2 = \int_0^1 \left[(v')^2 + v^2\right] \mathrm{d}x = \|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2, \tag{1.246}$$

$$B(u,v) = \int_0^1 u'v' \,\mathrm{d}x, \qquad L(v) = \int_0^1 fv \,\mathrm{d}x. \tag{1.247}$$

Check conditions.

1) **Symmetric:** $(u', v') = (v', u')$. ✓

2) **Continuous / bounded:** $|B(u,v)| \le \|u'\|_{L^2(I)} \|v'\|_{L^2(I)} \le \|u\|_V \|v\|_V$, so $\beta^* = 1$. ✓

3) **V-elliptic:** $B(v,v) = \|v'\|_{L^2(I)}^2 \ge \frac{1}{2}\left(\|v'\|_{L^2(I)}^2 + \|v\|_{L^2(I)}^2\right) = \frac{1}{2}\|v\|_V^2$, using **Poincaré** $\|v\|_{L^2(I)} \le \|v'\|_{L^2(I)}$ for $v \in H_0^1(I)$; so $\beta_* = \frac{1}{2}$. ✓

4) **L continuous:** $|L(v)| \le \|f\|_{L^2(I)} \|v\|_{L^2(I)} \le \|f\|_{L^2(I)} \|v\|_V$, so $\widetilde{L} = \|f\|_{L^2(I)}$. ✓

$\Rightarrow$ existence, uniqueness and boundedness, and Céa's lemma gives $\|u - u_h\|_V \le 2\|u - v\|_V$ for all $v \in V_h$, hence

$$\text{CG(1)}: \ \|u - u_h\|_V \le c_1 \|h\, u''\|_{L^2(I)}, \qquad \text{CG(2)}: \ \|u - u_h\|_V \le c_2 \|h^2 u'''\|_{L^2(I)}. \tag{1.248}$$

Intuition: Poincaré is what makes (VE) true

(VE) asks $B(v,v) \ge \beta_* \|v\|_V^2$, but $B(v,v)$ only sees v', while $\|v\|_V^2$ also contains v itself. Those are only comparable because of Poincaré, which says a function pinned to zero at the boundary cannot be large without having a large derivative. Remove the boundary condition — a free–free bar — and Poincaré fails for constants, $\beta_* = 0$, and $\boldsymbol{A}$ becomes singular. The rigid-body mode shows up as the failure of exactly this one inequality.

Part III — two dimensions

The 2D Poisson equation

$$-\Delta u = -u_{,xx} - u_{,yy} = f \quad \text{on } \Omega \subset \mathbb{R}^2, \qquad u = 0 \text{ on } \partial\Omega. \tag{1.249}$$

Multiply by a test function v and integrate over Ω:

$$-\iint_\Omega v\,\Delta u \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y. \tag{1.250}$$

Note (product rule). $\nabla\cdot(v\nabla u) = \nabla v\cdot\nabla u + v\,\Delta u$, so

$$-\iint_\Omega v\,\Delta u \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega \nabla u\cdot\nabla v \,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y. \tag{1.251}$$

By the **divergence theorem**,

$$\iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial\Omega} v\,\nabla u\cdot\boldsymbol{n}\,\mathrm{d}s, \qquad \boldsymbol{n} = \text{outward unit normal to } \partial\Omega. \tag{1.252}$$

Assuming $v = 0$ on $\partial\Omega$, the boundary integral vanishes and

$$\iint_\Omega \nabla u\cdot\nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y. \tag{1.253}$$

So

$$V = H_0^1(\Omega) = \left\{ v : \iint_\Omega \left[v^2 + \|\nabla v\|^2\right]\mathrm{d}x\,\mathrm{d}y < \infty,\ \ v = 0 \text{ on } \partial\Omega \right\}, \qquad \|\nabla v\|^2 = (v_{,x})^2 + (v_{,y})^2, \tag{1.254}$$

$$B(u,v) \coloneqq \iint_\Omega \nabla u\cdot\nabla v \,\mathrm{d}x\,\mathrm{d}y, \qquad L(v) \coloneqq \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y. \tag{1.255}$$

Check conditions.

1) **Symmetric:** $B(u,v) = B(v,u)$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|\nabla u\|_{L^2(\Omega)}\|\nabla v\|_{L^2(\Omega)} \le \|u\|_V\|v\|_V$, so $\beta^* = 1$. ✓
3) **V-elliptic:** $B(v,v) = \|\nabla v\|^2_{L^2(\Omega)} \ge \frac{1}{2}\big(\|\nabla v\|^2_{L^2(\Omega)} + \|v\|^2_{L^2(\Omega)}\big)$, using the **multi-dimensional Poincaré inequality**

$$\|v\|_{L^2(\Omega)} \le C\,|v|_{H^1(\Omega)} = C\|\nabla v\|_{L^2(\Omega)} \qquad \forall\, v \in H_0^1(\Omega), \tag{1.256}$$

giving $B(v,v) \ge \frac{1}{2}\min\big(\frac{1}{C^2},1\big)\|v\|_V^2$, i.e. $\beta_* = \frac{1}{2}\min\big(\frac{1}{C^2},1\big)$, a constant depending only on $\partial\Omega$. ✓

4) L **continuous:** $|L(v)| \le \|f\|_{L^2(\Omega)}\|v\|_{L^2(\Omega)} \le \|f\|_{L^2(\Omega)}\|v\|_V$, so $\widetilde{L} = \|f\|_{L^2(\Omega)}$. ✓

Intuition: nothing in Part II needed changing

The 2D problem was checked against the *same* four lines as the 1D problem, and passed. That is the return on the abstraction: Lax–Milgram gives existence and uniqueness, the SPD claim gives the matrix structure, and Céa gives the error bound — none of them re-proved. The only genuinely two-dimensional work was the divergence theorem in (1.252) and finding the right Poincaré constant.

CG(1) in 2D: the triangulation

Mesh $\mathcal{T}_h$: subdivide Ω into a finite number of triangles K_j, $j = 1, 2, \ldots, M$, with

$$h_j := \text{longest edge of } K_j, \tag{1.257}$$

$$h := \max_{1\le j\le M} h_j, \tag{1.258}$$

$$\rho_j := \text{diameter of the largest circle inscribed in } K_j. \tag{1.259}$$

Defn. $\mathcal{T}_h$ is a **regular** triangulation if there exists a constant $\alpha > 0$, *independent of* h, such that

$$\frac{h_j}{\rho_j} \le \alpha \qquad \forall\, j. \tag{1.260}$$

The discrete space is

$$V_h = \left\{ v \in H^1(\Omega) :\ v\big|_{K_j} = a_j + b_j x + c_j y,\ \ v\big|_{\partial\Omega} = 0 \right\}, \tag{1.261}$$

and Céa's lemma gives the error estimate

$$\|u - u_h\|_{H^1(\Omega)} \le c\, \|h\, u\|_{H^2(\Omega)}, \qquad |v|^2_{H^2(\Omega)} = \iint_\Omega \left[(v_{,xx})^2 + 2(v_{,xy})^2 + (v_{,yy})^2 \right] \mathrm{d}x\, \mathrm{d}y. \tag{1.262}$$

In mechanics terms: what regularity (1.260) is forbidding

h_j/ρ_j is an aspect ratio: longest edge over inscribed-circle diameter. Bounding it uniformly forbids *sliver* elements — long thin triangles that collapse towards a line as the mesh is refined. This is the same element-quality check a commercial mesher reports, and the reason it matters is that the interpolation estimate behind (1.262) carries $1/\rho_j$: let the triangles go slivery and the constant c blows up even though h is going to zero.

The beam, in the abstract language

The 1D fourth-order problem written as an abstract variational problem:

$$u^{(iv)} = f \ \text{ in } (0,1), \qquad u(0) = u(1) = u'(0) = u'(1) = 0, \qquad V = H^2_0(I),\ \ I = (0,1), \tag{1.263}$$

$$\|v\|_V = \|v\|_{H^2(\Omega)[I]} = \left(\int_0^1 \left[v^2 + (v')^2 + (v'')^2 \right] \mathrm{d}x \right)^{1/2}, \tag{1.264}$$

$$B(u,v) = (u'', v''), \qquad L(v) = (f, v). \tag{1.265}$$

Here $H^2_0(I)$ means $u = 0$ *and* $u' = 0$ on the boundary.
Check conditions.

1) **Symmetric:** $(u'', v'') = (v'', u'')$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|u''\|_{L^2(I)} \|v''\|_{L^2(I)} \le \|u\|_V \|v\|_V$, $\beta^* = 1$. ✓
3) **V-elliptic:** $B(v,v) = \|v''\|^2_{L^2(I)}$, and applying Poincaré twice, $\|v\|_{L^2(I)} \le \|v'\|_{L^2(I)} \le \|v''\|_{L^2(I)}$ for $v \in H^2_0(I)$, gives $B(v,v) \ge \frac{1}{3}\left(\|v\|^2_{L^2} + \|v'\|^2_{L^2} + \|v''\|^2_{L^2}\right) = \frac{1}{3}\|v\|^2_V$, so $\beta_* = \frac{1}{3}$. ✓
4) **L continuous:** $\widetilde{L} = \|f\|_{L^2(0,1)}$. ✓

CG(3) estimate (Céa's lemma):

$$\|u - u_h\|_{H^2(\Omega)[(0,1)]} \le c\, \|h^2 u\|_{H^4(\Omega)[(0,1)]}, \qquad |v|^2_{H^4(\Omega)[(0,1)]} = \int_0^1 \left(v''''\right)^2 \mathrm{d}x. \tag{1.266}$$

Recall: this is HW3 Problems 4–5, done abstractly

The homework proved $\|u - U\|_E = O(h^2)$ by hand, with Galerkin orthogonality, Cauchy and the interpolation estimate. (1.266) is the same result read straight off Céa's lemma with $\beta^*/\beta_* = 3$. The hand proof gave a better constant because it used the energy norm, in which $\beta^* = \beta_*$; the abstract route gives the same *rate* with less work.

The biharmonic equation [12/Oct/26]

The 2D plate — the beam's two-dimensional counterpart:

$$\Delta^2 u = \Delta(\Delta u) = f \implies u_{,xxxx} + 2u_{,xxyy} + u_{,yyyy} = f, \tag{1.267}$$

$$\text{BCs:} \qquad u\big|_{\partial\Omega} = 0, \qquad \nabla u \cdot \boldsymbol{n}\big|_{\partial\Omega} = 0. \tag{1.268}$$

Multiply by v and integrate, letting $z = \Delta u$:

$$\iint_\Omega v\,\Delta(\Delta u)\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega v f\,\mathrm{d}x\,\mathrm{d}y,$$

$$\iint_\Omega v\,\Delta z\,\mathrm{d}x\,\mathrm{d}y = -\iint_\Omega \nabla v\cdot\nabla z\,\mathrm{d}x\,\mathrm{d}y + \oint_{\partial\Omega} v\,\nabla z\cdot\boldsymbol{n}\,\mathrm{d}s, \tag{1.269}$$

and taking $v = 0$, $\nabla v\cdot\boldsymbol{n} = 0$ on $\partial\Omega$,

$$B(u,v) = \iint_\Omega \Delta u\,\Delta v\,\mathrm{d}x\,\mathrm{d}y, \qquad L(v) = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y, \qquad V = H^2_0(\Omega). \tag{1.270}$$

Note. If $v \in H^2_0(\Omega)$ then

$$\|v\|^2_{H^2(\Omega)} = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)} + \|v_{,xx}\|^2_{L^2(\Omega)} + 2\|v_{,xy}\|^2_{L^2(\Omega)} + \|v_{,yy}\|^2_{L^2(\Omega)}, \tag{1.271}$$

and $v \in H^2_0(\Omega)$ means $v = 0$, $v_{,x} = 0$, $v_{,y} = 0$ on $\partial\Omega$.
Check conditions.

1) **Symmetric:** $B(u,v) = B(v,u)$. ✓
2) **Continuous / bounded:** $|B(u,v)| \le \|\Delta u\|_{L^2(\Omega)}\|\Delta v\|_{L^2(\Omega)} \le \|u\|_V\|v\|_V$. ✓
3) **V-elliptic:** $B(v,v) = \|\Delta v\|^2_{L^2(\Omega)}$, and by a result of **Girault–Raviart (1981)**,

$$\|v\|_{H^2(\Omega)} \le C\,\|\Delta v\|_{L^2(\Omega)} \qquad \text{for } v \in H^2_0(\Omega), \tag{1.272}$$

hence $B(v,v) \ge \frac{1}{C^2}\|v\|^2_V$. ✓
4) **L continuous:** $\widetilde{L} = \|f\|_{L^2(\Omega)}$. ✓

Conforming FEM for the biharmonic: the Argyris element

A conforming method needs $V_h \subset V = H^2_0(\Omega)$, i.e.

$$V_h = \Big\{ v \in H^2(\Omega) :\ v\big|_{K_j} = \sum_{\ell+m=0}^{\ell+m\le 5} a_j^{(\ell m)}\, x^{(\ell)} y^{(m)},\ \ v\big|_{\partial\Omega} = 0,\ \ \nabla v\cdot\boldsymbol{n}\big|_{\partial\Omega} = 0 \Big\}, \tag{1.273}$$

a complete quintic in two variables — **21 different polynomials** on each triangle. The corresponding element is the **Argyris element (1968)**, whose 21 degrees of freedom per triangle are

$$\underbrace{u,\ \nabla u,\ u_{,xx},\ u_{,xy},\ u_{,yy}}_{\text{6 per vertex} \times \text{3 vertices} = 18} \qquad \text{plus} \qquad \underbrace{\nabla u\cdot\boldsymbol{n}}_{\text{1 per edge} \times \text{3 edges} = 3}. \tag{1.274}$$

In mechanics terms: why plates are hard, in one element

A conforming plate element must be C^1 across element boundaries — the 2D version of the beam's requirement — and matching *both* the value and the full gradient along a shared edge is far more demanding than matching them at a point. The cheapest complete polynomial that can do it is a quintic, which is why the Argyris element carries second derivatives as nodal unknowns and ends up with 21 DOFs on a single triangle.
This is also why real plate and shell codes usually do *not* use conforming elements: they drop to non-conforming or mixed formulations (and give up $V_h \subset V$) rather than pay this price.

Part IV — 2D mesh generation

Triangulations

Note. FEM has no inherent restriction on the mesh. The simplest and most flexible choice is to use a **simplex**: 2D triangles, 3D tetrahedra.

Note. A general triangulation in 2D is **unstructured**, meaning grid points do not lie along coordinate lines.

Defn (triangulation). Let Ω be a 2D domain with polygonal boundary $\partial\Omega$. A triangulation $\mathcal{T}_h = \{K_j\}_{j=1}^{M}$ is a subdivision of Ω into M non-overlapping triangles constructed so that no vertex of one triangle lies on the edge of another triangle:

$$\bigcup_{j=1}^{M} K_j = \Omega. \tag{1.275}$$

Let $X_h = \{x_i = (x_i, y_i)\}_{i=1}^{N}$ denote the set of **nodes** (vertices of the triangles) and $\mathcal{S}_h = \{s_k\}_{k=1}^{E}$ the **edges**.

Main question (Mesh Generation, Part 1). Given a collection of points in $\mathbb{R}^2$ (nodes), how can we connect them to form a triangulation?

Voronoi diagrams

Idea. Given a finite set of points in $\mathbb{R}^2$, assign to each point a *region of influence* in such a way that it decomposes $\mathbb{R}^2$.

Defn (Voronoi region). Let X be a collection of N points in $\mathbb{R}^2$. For each $x_i \in X$ the **Voronoi region** V_i is the set of points $x \in \mathbb{R}^2$ that are at least as close to x_i as to any other point in X:

$$V_i = \left\{x \in \mathbb{R}^2 : \ \|x - x_i\| \leq \|x - x_j\| \ \ \forall\, x_j \in X\right\}. \tag{1.276}$$

Consider the half-plane

$$H_{ij} = \left\{x \in \mathbb{R}^2 : \ \|x - x_i\| \leq \|x - x_j\|\right\}. \tag{1.277}$$

The Voronoi region of x_i is the convex intersection of the half-planes H_{ij} for all $j \neq i$; therefore V_i is a convex polygonal region (possibly unbounded).

Notes.

(1) Each point in $\mathbb{R}^2$ has at least one nearest point in X, so every $x \in \mathbb{R}^2$ lies in at least one Voronoi region.

(2) Two Voronoi regions lie on opposite sides of a perpendicular bisector; two Voronoi regions do not share any interior points. If a point belongs to more than one region, it must lie on the boundary.

(3) Voronoi regions for a given X cover $\mathbb{R}^2$.

(4) Voronoi regions together with edges and vertices form the **Voronoi diagram** of X.

(5) For a given X, the Voronoi diagram is unique.

Delaunay triangulation

Defn (Delaunay partition). Let $X \subset \mathbb{R}^2$ be a set of N points and let V be the Voronoi diagram. The **Delaunay partition** is obtained by creating an edge between two points in X *if and only if* their Voronoi regions share a boundary.

Generic situation: 3 points of X share a common vertex, so a triangle is formed.

Degenerate situation: 4 or more points share a common Voronoi vertex — i.e. they are co-circular.

Note. Degeneracies can always be converted to a triangulation (a square is split by either diagonal).

Defn (Delaunay triangulation). The Delaunay triangulation $\mathcal{T}_h$ is the Delaunay partition together with some strategy to convert degenerate elements into triangles.

Properties.

(1) The Voronoi diagram and the Delaunay triangulation form a **duality**.

(2) For a given $X \subset \mathbb{R}^2$, V is unique, and $\mathcal{T}_h$ is unique up to degeneracies.

(3) $\mathcal{T}_h$ contains $O(N)$ triangles.

(4) $\mathcal{T}_h$ **maximises the minimum angle**.

(5) A circle circumscribing any triangle in $\mathcal{T}_h$ does not contain any other point of X.

In mechanics terms: property (4) is the one you care about

"Maximises the minimum angle" is exactly the regularity condition (1.260) in disguise: of all possible triangulations of the same point set, Delaunay is the one that comes closest to avoiding slivers. That is why essentially every unstructured mesher starts from a Delaunay triangulation — it is not merely convenient, it is optimal for the constant in the interpolation estimate.
Property (5), the empty-circumcircle condition, is the practical test a code uses to check or repair a mesh.

In Python

1) Create a list of points in $\mathbb{R}^2$: `p[0:N, 0:2]`, where the first index is the point index and the second the coordinate (x, y).

2) Create $\mathcal{T}_h$ (Delaunay triangulation):

```
import scipy.spatial as sp
t = sp.Delaunay(p).simplices
```

3) Plot the mesh:

```
import numpy as np
import matplotlib.pyplot as plt
plt.triplot(p[:,0], p[:,1], t, facecolors=np.zeros(N), edgecolors='k')
```

4) Plot a function defined on the nodes:

```
U = ufunc(p)
plt.tripcolor(p[:,0], p[:,1], t, U, shading='flat', cmap='inferno')
```

What is `t`? N = number of nodes, M = number of triangles; `t` is an $M \times 3$ array of integers, where `t[j][i]` gives, for triangle j, the node index $i \in \{0, 1, 2\}$.

Note (area of a triangle). For vertices (x_0, y_0), (x_1, y_1), (x_2, y_2), set $\underline{v}_1 = (x_1 - x_0,\ y_1 - y_0,\ 0)$ and $\underline{v}_2 = (x_2 - x_0,\ y_2 - y_0,\ 0)$; then

$$\text{area} = \tfrac{1}{2}\left\|\underline{v}_1 \times \underline{v}_2\right\|. \tag{1.278}$$

Note. Delaunay in C: `http://www.qhull.org`.

Mesh generation, Part 2: representing Ω and $\partial\Omega$

Idea. Describe the domain *implicitly* using a **level-set function**.

Ex. $\phi(x) = \|x\| - R$, where $R > 0$. Then

$$\partial\Omega = \left\{x \in \mathbb{R}^2 :\ \phi(x) = 0\right\} = \left\{x \in \mathbb{R}^2 :\ \|x\| = R\right\}, \qquad \Omega = \left\{x \in \mathbb{R}^2 :\ \phi(x) < 0\right\}, \tag{1.279}$$

a circle of radius R centred at $(0, 0)$.

Note (convention).

$$\{\phi = 0\} \text{ is the boundary}, \qquad \{\phi < 0\} \text{ is the interior of } \Omega, \qquad \{\phi > 0\} \text{ is the exterior of } \Omega. \tag{1.280}$$

Note. For a given $\partial\Omega$ there are many level-set functions; e.g. $\phi(x) = \|x\|^p - R^p$ for any positive integer p describes the disc of radius R centred at $(0, 0)$.

Boolean operations.

$$\phi = \min(\phi_1, \phi_2) \qquad \text{is the } \textbf{union} \text{ operation: } \Omega = \Omega_1 \cup \Omega_2, \tag{1.281}$$

$$\phi = \max(\phi_1, \phi_2) \qquad \text{is the } \textbf{intersection} \text{ operation: } \Omega = \Omega_1 \cap \Omega_2. \tag{1.282}$$

Intuition: why min is union

With the sign convention (1.280), "inside" means negative. A point is in $\Omega_1 \cup \Omega_2$ when it is inside *at least one* of them, i.e. when *at least one* ϕ_i is negative — which is exactly $\min(\phi_1, \phi_2) < 0$. Intersection needs *both* negative, hence max. CAD solid modelling is built on these two lines.

Translation: mesh generation

Math department writes	You should hear
level-set function ϕ	implicit CAD geometry; $\phi < 0$ is material
Voronoi region	the territory nearest a given node
Delaunay triangulation	the dual mesh; the "best" triangles for a point set
$h_j / \rho_j \le \alpha$	element aspect-ratio / sliver check
`scipy.spatial.Delaunay`	what a mesher calls underneath

► **Unclear in my notes:** The scan's `plt.triplot` call is written with both `facecolors` and `edgecolors` arguments, which belong to `tripcolor`, not `triplot`. Transcribed as written; worth checking the lecturer's actual code before relying on it.